INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON



Réseaux de capteurs et IoT: Architectures, protocoles et sécurité

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citi lab

Inria INVENTEURS DU MONDE NUMÉRIQUE

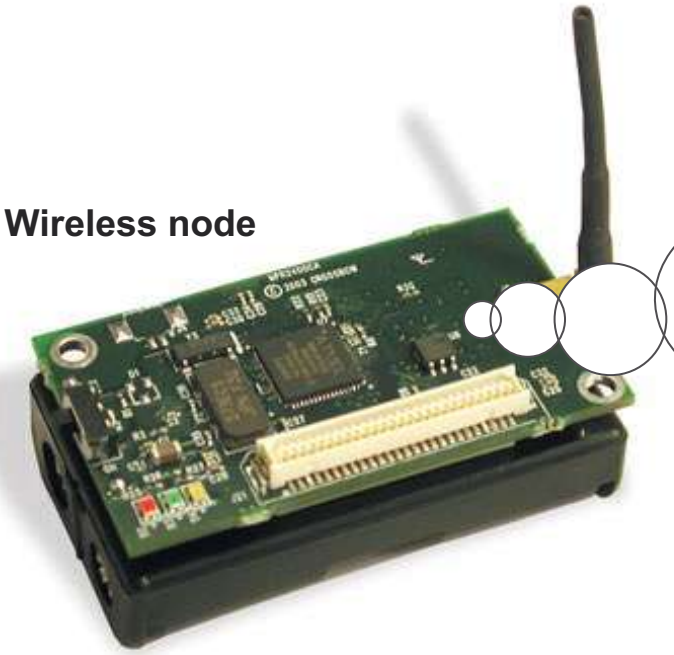
membre de UNIVERSITÉ DE LYON

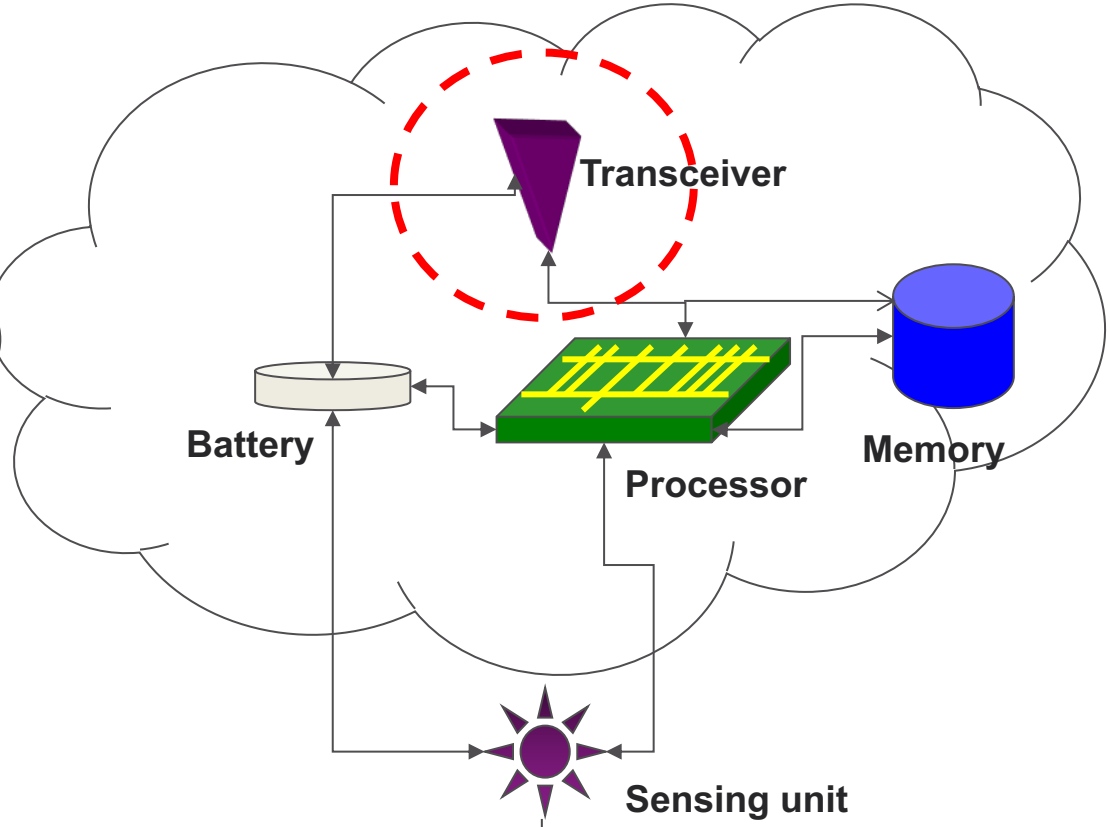
1.1 Internet des Objets et Réseaux de Capteurs Sans Fil

Contexte général

Architecture d'un nœud capteur

Wireless node





Example of external sensors



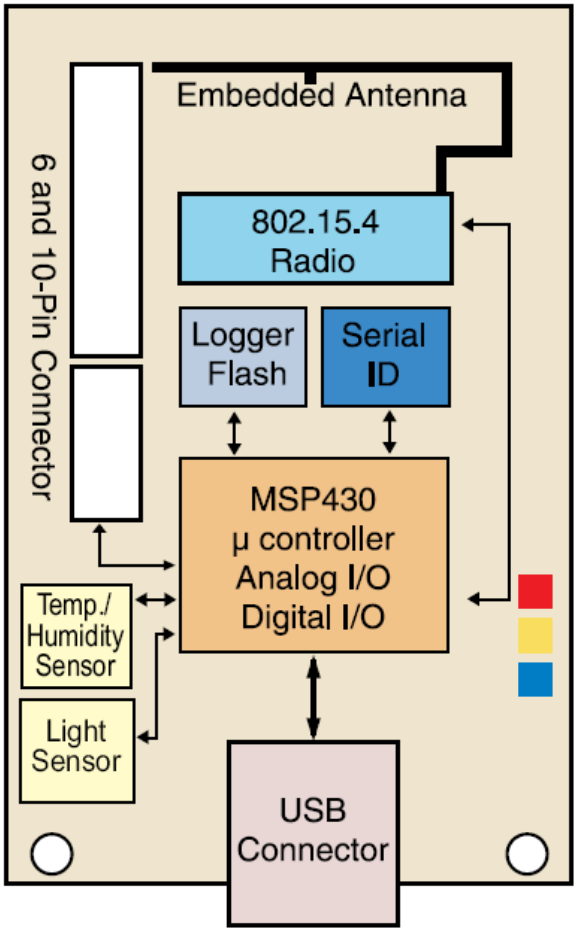
Exemple de Telosb

- **Processeur**
  - –TI MSP430
  - –8 MHz
  - –10kB RAM
- **Transmission**

IEEE 802.15.4
(**TI CC2420**)

  - 250 Kbps (2.4 - 2.4835 GhZ Band)
  - Embedded antenna
- **Flash**
  - –1 MB
- **Sensors**

  Light, Temperature and Humudity



Carte MKR 1010

- Alimentation (port micro-USB, batterie LiPo 3,7, broche VIN= 5v)
- Microcontrôleur: SAMD21 Cortex M0+, 32bits à 48 MHz
- Mémoire flash: 256 kB
- Mémoire SRAM: 32 kB
- Tension de sortie: 3,3 Vcc

- **Secure element (ATECC508A)**

- 7 entrées analogiques ADC 8, 10 (par défaut) et 12 bits
- 1 sortie analogique DAC 10 bits

- 8 E/S numériques
- SPI (MOSI, SCK, MISO)
- I2C (SCL,SDA)
- UART (TX, RX)

+ WiFi & Bluetooth
(u-blox NINA-W102)



Carte MKR 1310

- Alimentation (port micro-USB, batterie LiPo 3,7, broche VIN= 5v)
- Microcontrôleur: SAMD21 Cortex M0+, 32bits à 48 MHz
- Mémoire flash: 256 kB
- Mémoire SRAM: 32 kB
- Tension de sortie: 3,3 Vcc



- **Secure element (ATECC508A)**

- 7 entrées analogiques ADC 8, 10 (par défaut) et 12 bits
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- 8 E/S numériques
- SPI (MOSI, SCK, MISO)
- I2C (SCL,SDA)
- UART (TX, RX)

- + LoRa / LoRaWAN (Murata CMWX1ZZABZ)

Applications de l'IoT

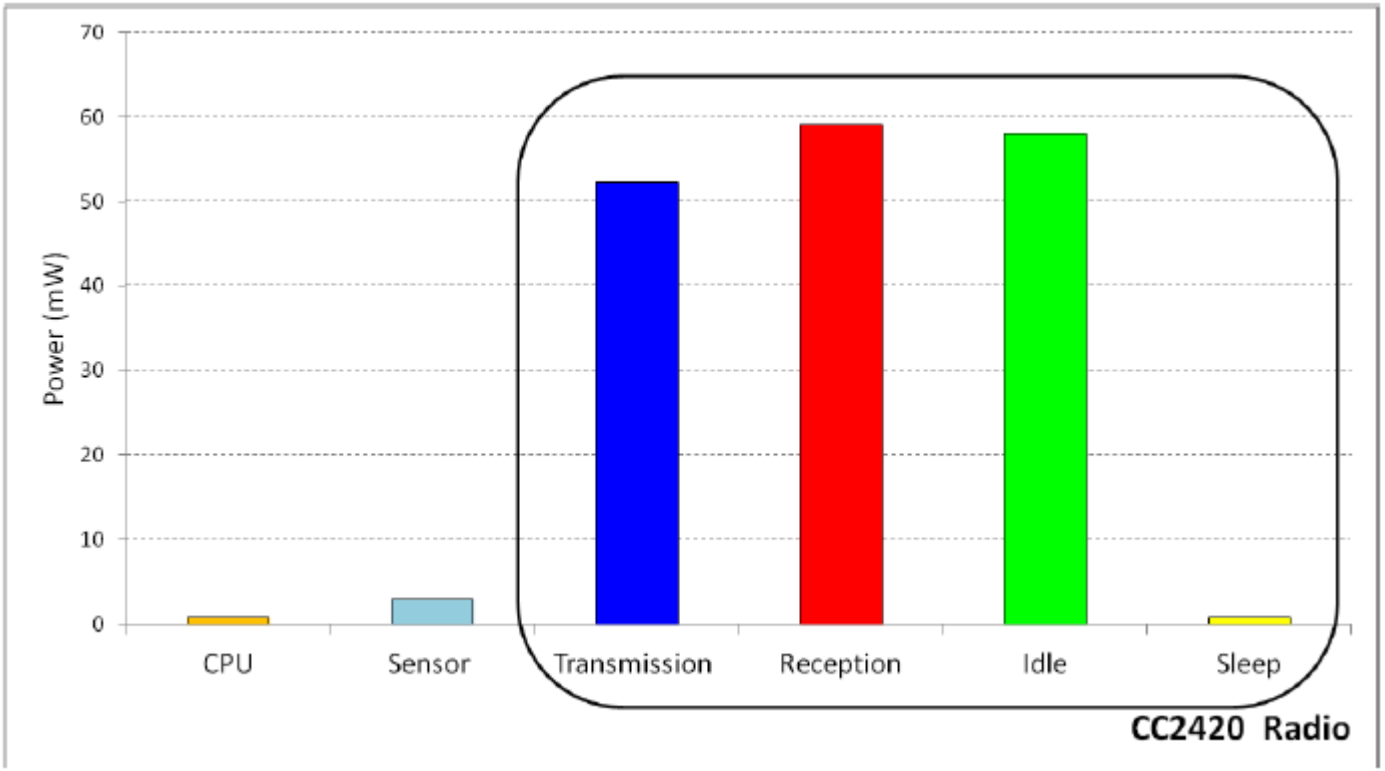


Caractéristiques des objects de l'IoT

- Limitation des ressources :
  - **Contraintes de consommation d'énergie**
  - **Contraintes de mémoire**
- Manque d'infrastructures
- Environnement sans fil
- Déploiement à grande échelle
- Couplage avec l'environnement physique
- Mobilité des nœuds
- Hétérogénéité des nœuds

Contraintes d'énergie dans les nœuds capteurs

- Limitation de ressources et principalement l'énergie
- La consommation de la radio est la plus importante



Security challenges in WSN

IT life cycle

- Factorized software
- Upgrades
- Soft restes when unusable
- Large scale
- Mature codebase

IoT life cycle

- Fragmentation
- No upgrades
- Non soft restes
- Very large scale
- Young codebase

How IoT revisit security problems, François Lesueur, INSA-Lyon, April 2017

+ It's not the walls that protect the citadel, but the spirit of its inhabitants."

[Thucydides]

At least 40% of attacks are caused by users themselves (ex. Mirai botnet).

Security challenges in WSN

Information Security

$$Risk = Threat \times Vulnerability \times Cost$$

Vulnerability

- Deployment environment
- Lack of physical protection
- Memory limitation
- Energy limitation
- Wireless Communication
- Lack of infrastructure

Threat

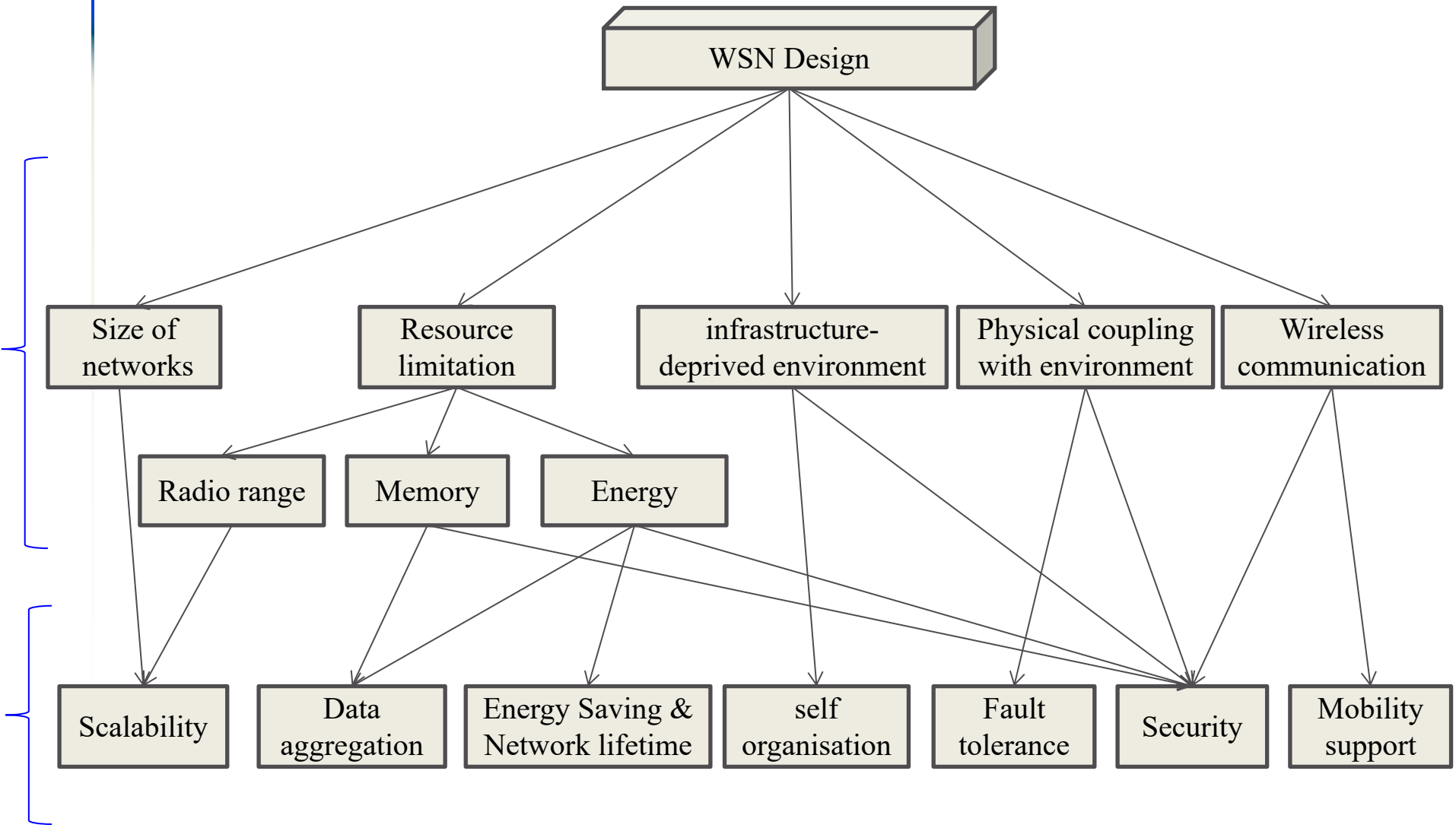
- Traditional threats (eavesdropping, message injection ...)
- New specific attacks (energy drain, data aggregation attacks)

Cost

- Critical applications of WSN
- Dramatic effects and costs

Ex: : Mirai botnet (DDoS attacks in 2016/17)

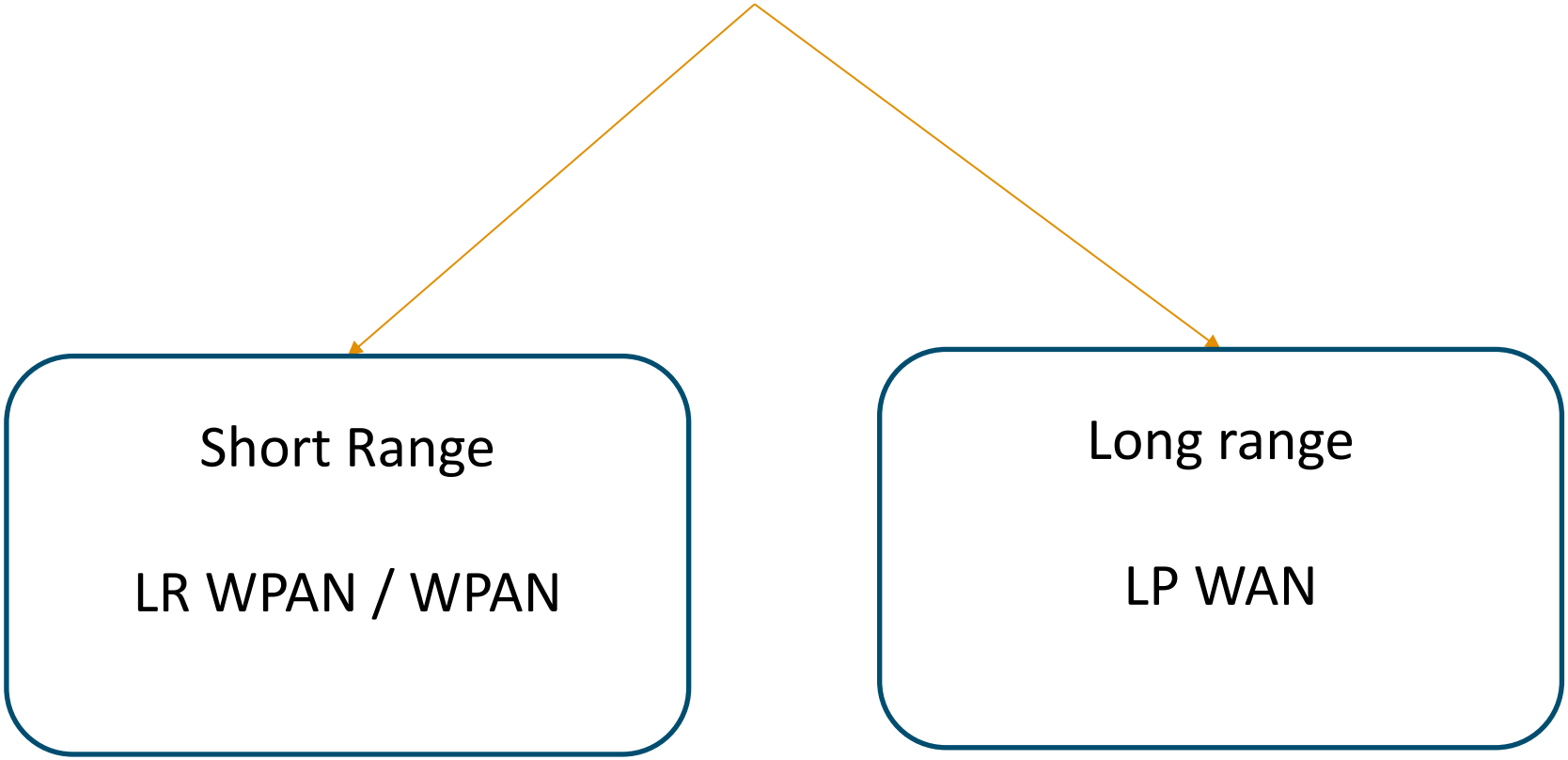
WSN: From characteristics to challenges



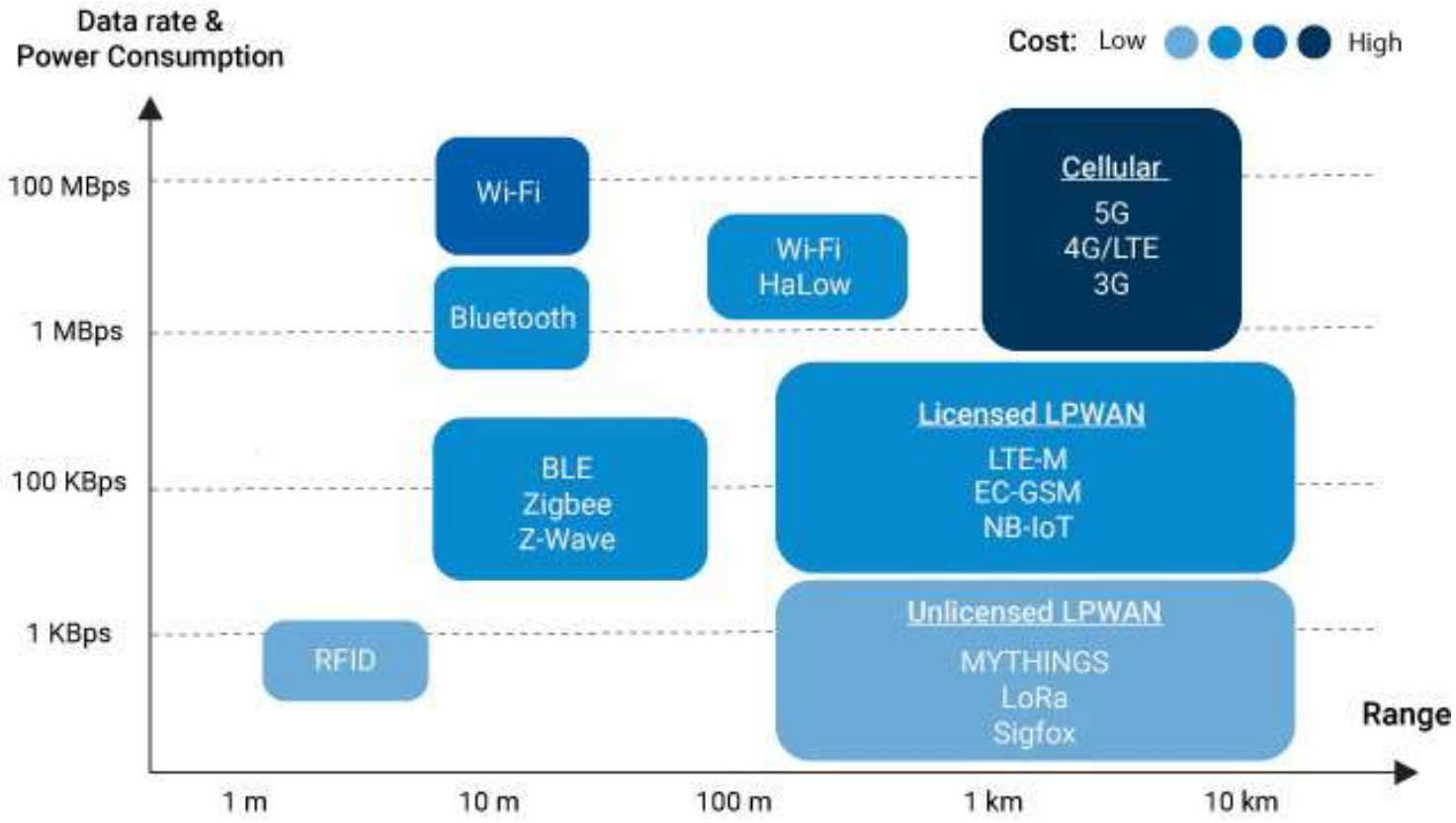
1.2 Protocoles de communication à faible consommation d'énergie

Solution standardisées

Protocoles sans fil à basse consommation d'énergie (pour l'IOT)

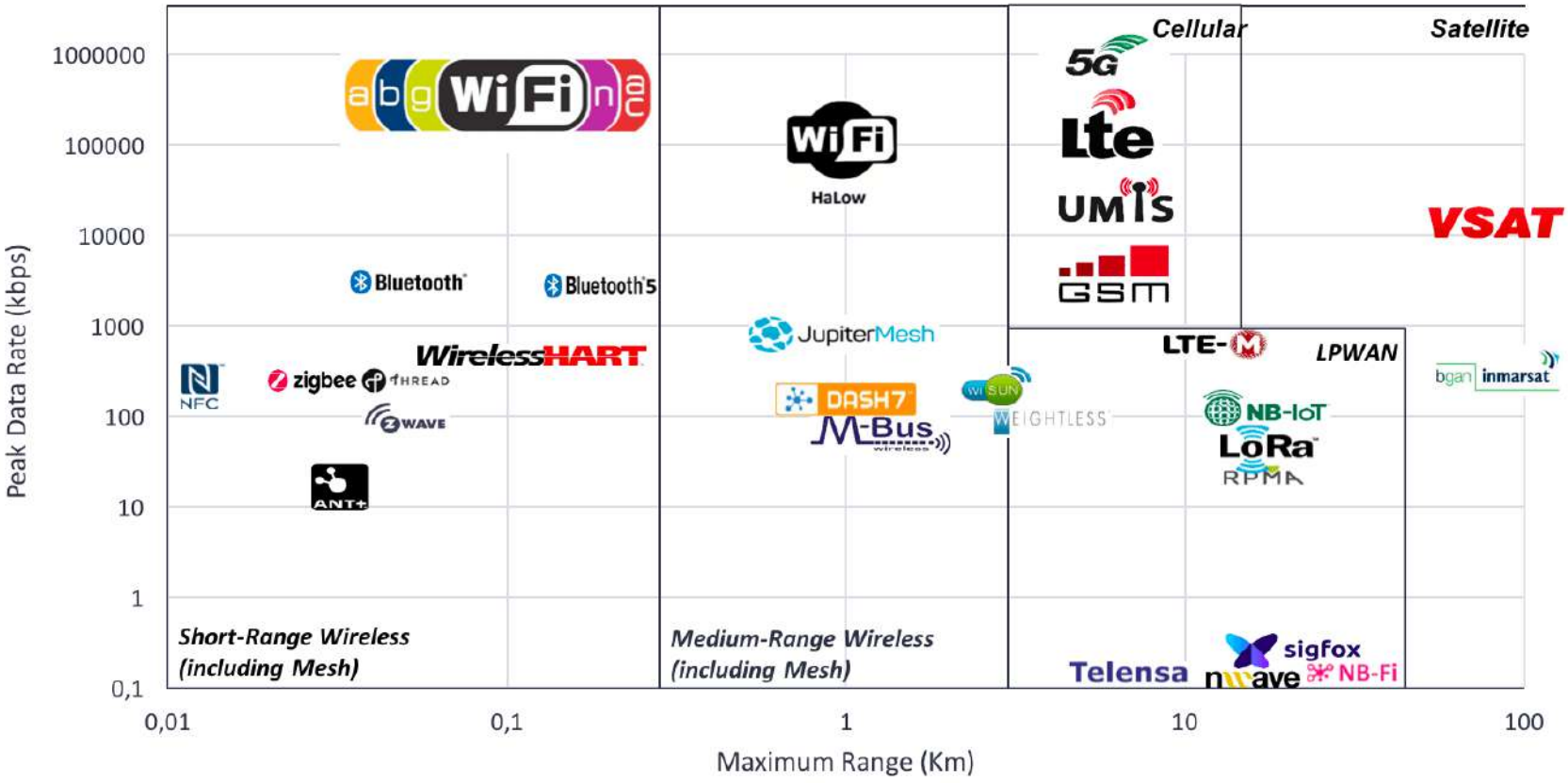


Classification des principaux standards



Source: https://behrtech.com/blog/6-leading-types-of-iot-wireless-tech-and-their-best-use-cases/

Classification des principaux standards



Source: https://iot-analytics.com/iot-segments/iot-connectivity/

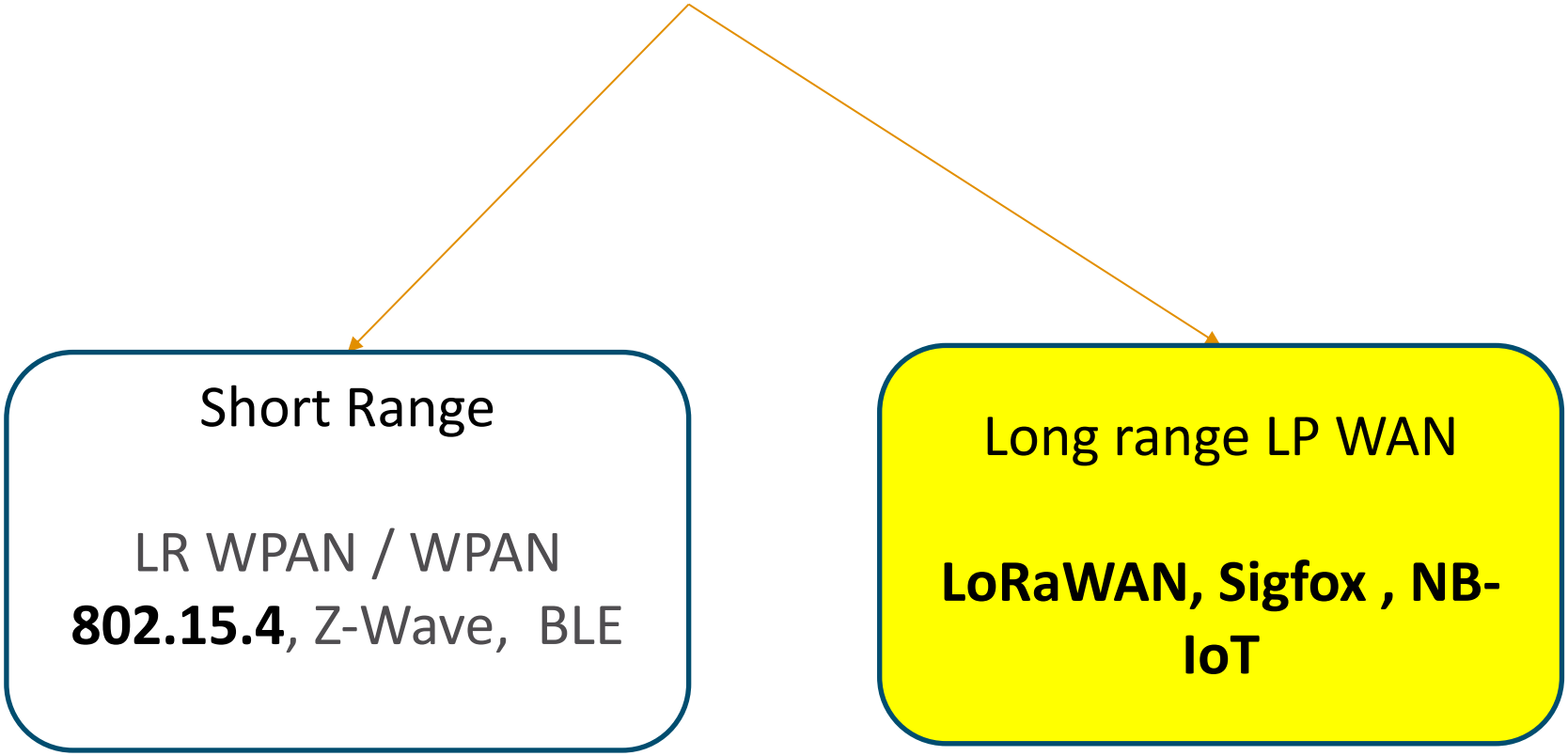
Classification des principaux standards

Attribute	Bluetooth® Low Energy Technology	Wi-Fi	Z-Wave	IEEE 802.15.4 (Zigbee, Thread)	LTE-M	NB-IoT	Sigfox	LoRaWAN
Range	10 m – 1.5 km	15 m – 100 m	30 m - 50 m	30 m – 100 m	1 km – 10 km	1 km – 10 km	3 km – 50 km	2 km – 20 km
Throughput	125 kbps – 2 Mbps	54 Mbps – 1.3 Gbps	10 kbps – 100 kbps	20 kbps – 250 kbps	Up to 1 Mbps	Up to 200 kbps	Up to 100 bps	10 kbps – 50 kbps
Power Consumption	Low	Medium	Low	Low	Medium	Low	Low	Low
Ongoing Cost	One-time	One-time	One-time	One-time	Recurring	Recurring	Recurring	One-time
Module Cost	Under $5	Under $10	Under $10	$8-$15	$8-$20	$8-$20	Under $5	$8-$15
Topology	P2P, Star, Mesh, Broadcast	Star, Mesh	Mesh	Mesh	Star	Star	Star	Star
Shipments in 2019 (millions)	~3,500	~3,200	~120	~420	~7	~16	~10	~45

Source: https://www.bluetooth.com/blog/wireless-connectivity-options-for-iot-applications-technology-comparison/

1.3 Solutions long range

Protocoles sans fil pour l'IOT



Fréquences sans licences: LoRa ou Sigfox

sigfox



LoRa™

LoRaWAN™

Long Range Low Power: SigFox / Lora

Sigfox (2009): appuyé par GDF Suez, Air liquide, Telefónica, etc.)

LoRa : Cycléo 2009 acheté par Semtech **en 2012.**

LoRa Alliance Semtech Orange, Objenious by Bouygues Telecom, STMicroelectronics, Cisco , IBM, etc.

Long Range Low Power: SigFox / Lora

	SigFox	LoRa / LoRaWAN
Bande ISM	Oui	Oui
Abonnement	Oui	Pas nécessairement
Largeur de bande (Khz)	192 Khz	125-500 Khz
Modulation	UNB	CSS
Débit (b/s)	100	~ 250 – 11000 (125Khz)
Portée maximale (km)	63	22
Taille max du payload (o)	12	255 (Lora), 51 à 222 en LoRaWAN
Type de réseau	Un opérateur unique	Réseau privé ou opéré
Uplink	140 messages / jour	Illimité (duty cycle ISM)
Downlink	4 messages / jour	Illimité (duty cycle ISM)

LPWAN: résumé de la situation en France

- Orange se focalise sur LoRa/LoRaWAN et LTE-M
- Bouygues via Objenious se focalise sur LoRa/LoRaWAN
- SFR se focalise sur NB-IoT
- SigFox : opérateur mondial des réseaux SigFox

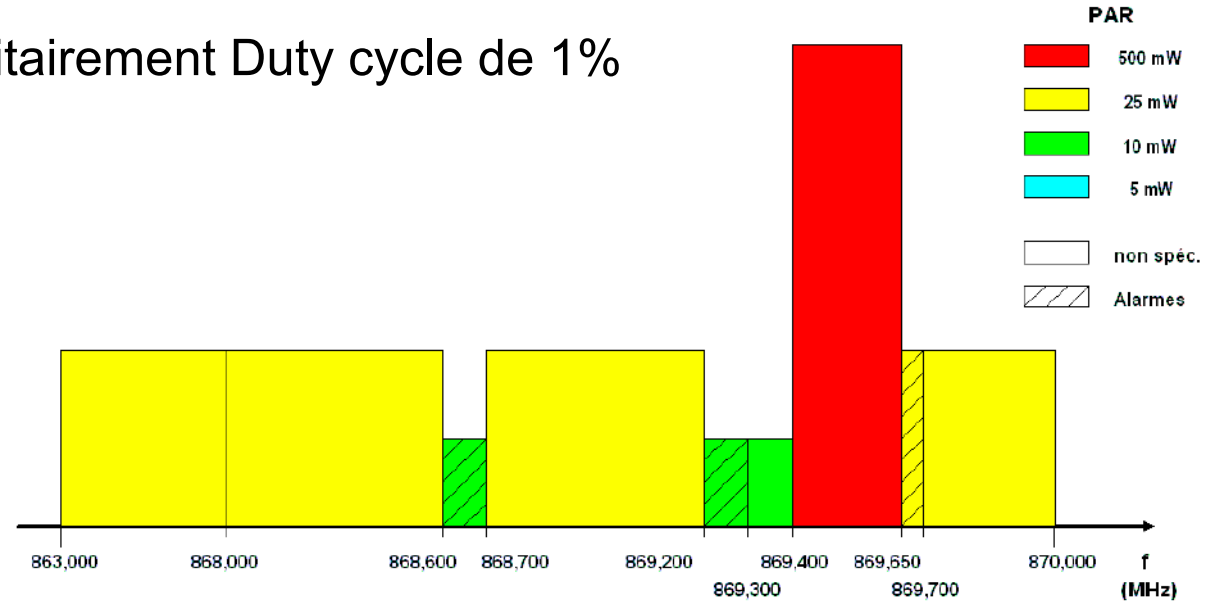
  → Reprise par UnaBiz en avril 2022

1.3 Solutions long range

Exemple: LoRa/LoRaWAN

LoRa: Fréquences

- Bandes des 433 ou 868 MHz (en Europe), 915 MHz (en Amérique)...
- En Europe: Puissances Apparentes Rayonnées limité à 10 mW sur toute la bande 433 MHz et variable sur la bande 868 MHZ
- Majoritairement Duty cycle de 1%



Source de la figure: http://www.onditech.com/documents/freq_433_868.pdf

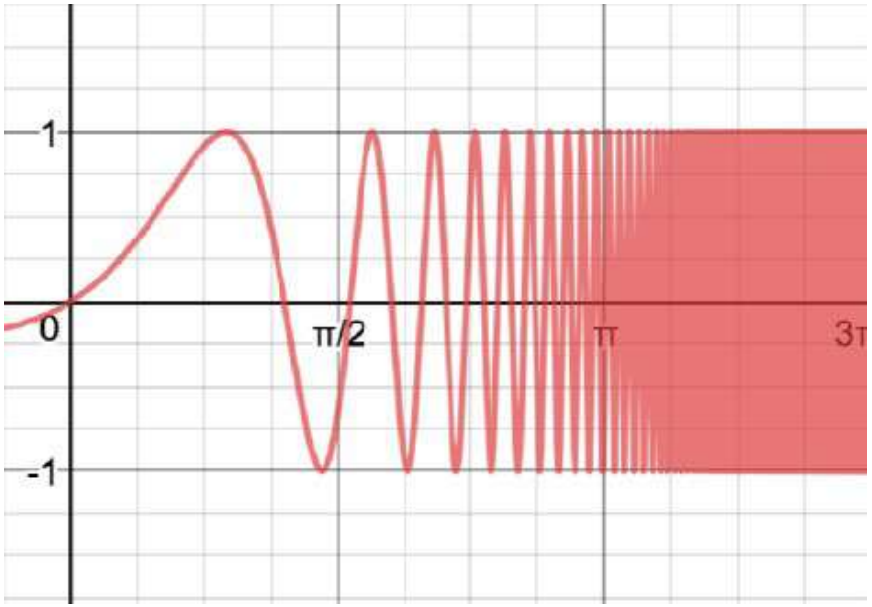
LoRa: Couche Physique

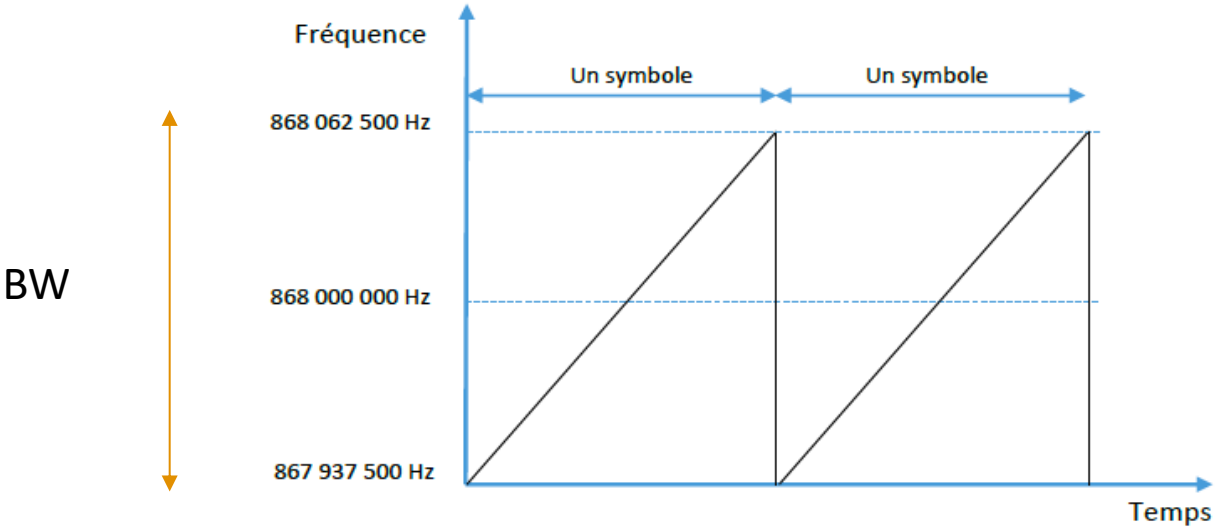
Spreading Factor (SF)	**Puissance d'émission, Sensibilité de réception**
Coding Rate (CR)	**Largeur de bande (Bandwidth)**

LoRa: Couche Physique (modulation)

- Modulation utilisé

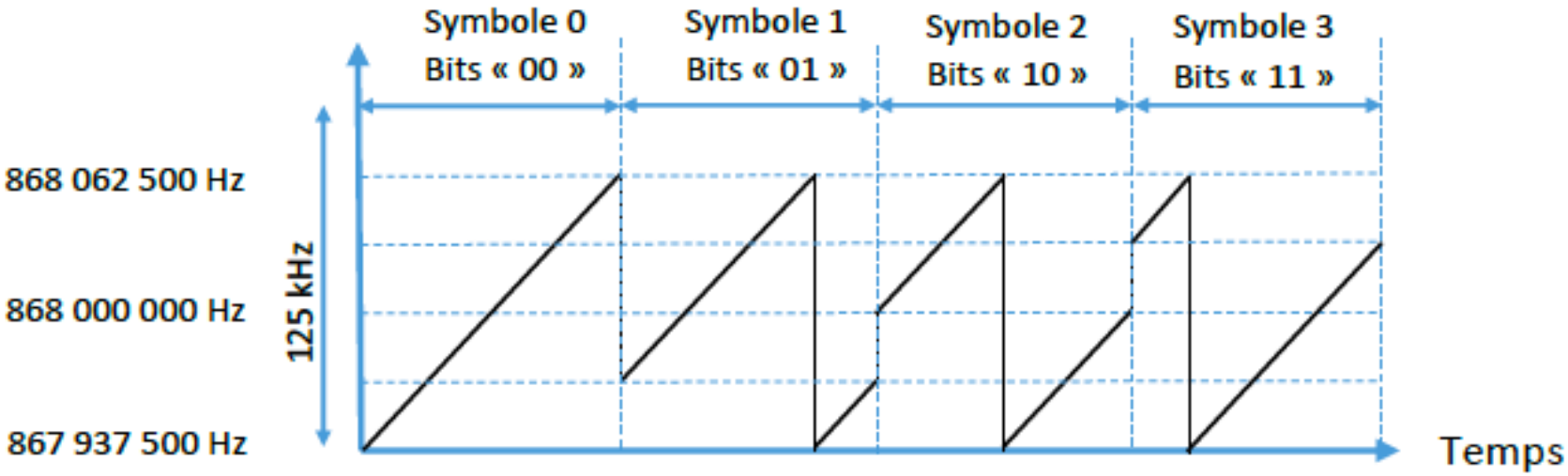
CSS (Chirp spread spectrum)





LoRa: Couche Physique (SF et modulation)

- SF : nombre de bits par symbole 1 Symbole = SF bits
- Chaque symbole est représenté par un Chirp
- On a donc 2^{SF} forme de symboles possibles (Chirp).
- Exemple : SF=2



Source: cours Fomation LoRa / LoRaWAN, Sylvain MONTAGNY, Université Savoie Mont Blanc

LoRa: Couche Physique (TX et Ps)

- **Puissance maximale en Europe 25 mW = 14 dBm**
- Le grand avantage de Lora est qu'on peut démoduler le signal même très faible (jusqu'à -120 à -140 dBm)
- On peut démoduler même avec un SNR compris entre -5dB et -20dB

SpreadingFactor (RegModemConfig2)	Spreading Factor (Chips / symbol)	LoRa Demodulator SNR
6	64	-5 dB
7	128	-7.5 dB
8	256	-10 dB
9	512	-12.5 dB
10	1024	-15 dB
11	2048	-17.5 dB
12	4096	-20 dB

Source: cours Fomation LoRa / LoRaWAN, Sylvain MONTAGNY, Université Savoie Mont Blanc

LoRa: Couche Physique (Bandwidh)

- Largeur de la bande

- Généralement les valeurs par défaut sont de 125Khz mais on utilise aussi 62.5Khz, 250 Khz, 250 Khz, 500 Khz

- En doublant la largeur de la bande, on divise le temps d'un symbole par deux, et on double le débit

LoRa: Couche Physique (C/R)

- Rappelez-vous du Forward Error correction, la redondance au niveau physique pour détecter et corriger les erreurs

- CR' = 4 / (4 + CR)
- CR = 1, 2 , 3 ou 4
- CR' = 4/5 , 4/6 , 4/7 ou 4/8

CR	CR'	Surplus
1	4/5	25%
2	4/6	50%
3	4/7	75%
4	4/8	100%

+ robuste (détection/ correction les erreurs)

+ de bits à envoyer

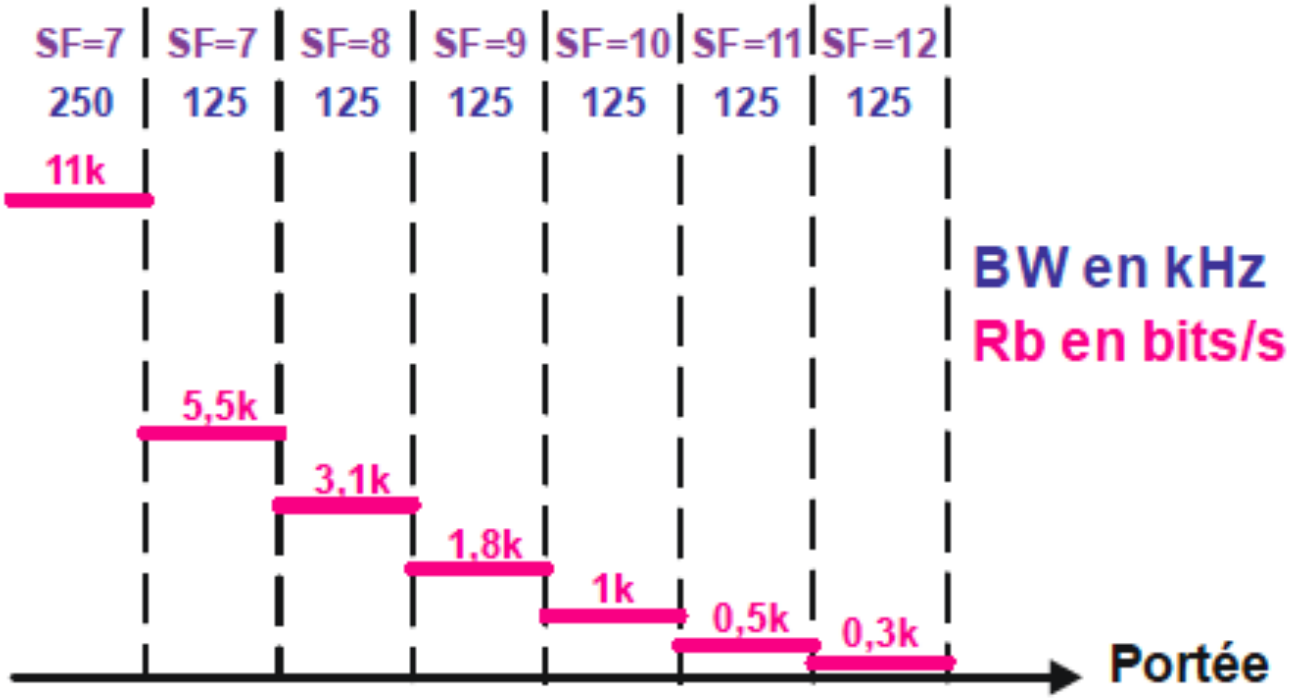
LoRa: Couche Physique (Débit utile)

Débit théorique en appliquant la formule

SF	BW (KHz)	CR 4/5	CR 4/8
7	250	10938	6836
7	125	5469	3418
8	125	3125	1953
9	125	1758	1099
10	125	977	610
11	125	537	336
12	125	293	183

- Attention à l'arrondi
- Les débits en pratique sont très proches (Voir TP ☺)

LoRa: Couche Physique (Débit utile CR4/5)

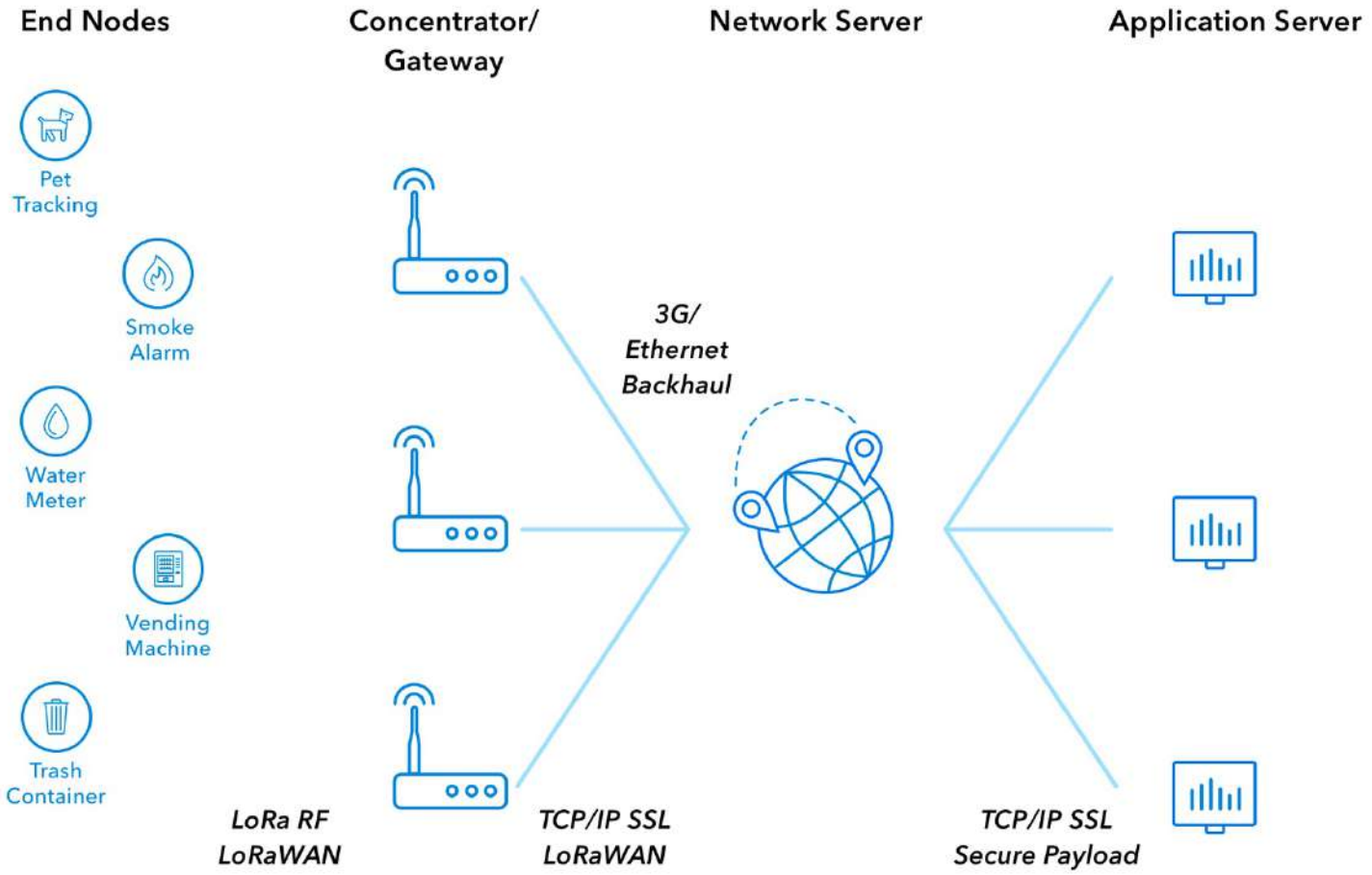


Source: Caractérisation des procédés de modulation et de démodulation LoRa, J-F SERREAU, 2019

LoRa: Résumé

Spreading Factor (SF)	**Puissance d'émission, Sensibilité de réception**
Coding Rate (CR)	**Largeur de bande (Bandwidth)**

LoRaWAN : Architecture



source: https://www.thethingsnetwork.org/

LoRaWAN : Architecture

- **Noeud (Device LoRaWaN):** transmission des données (souvent issues de capteurs) et réception des données en voix descendante.

- **Passerelle (Gateway LoRaWAN):**

  - pont entre les nœuds et le réseau cœur (Serv. réseau et application).

  - Ecoute sur plusieurs canaux LoRa pour tous les SF à la fois

- **Serveur Réseau (Network server)**:

  - Intégrité des données, dé-duplication des messages

  - lien avec le serv. applicatif, choix de passerelle en voix descendante.

- **Serveur Applicatif:**

  - Déchiffrer les messages et les presenter aux utilisateurs

LoRaWAN : Identifiants et clés à connaître

- **Identifiants à connaitre:**
  - DevEUI - 64 bit identifiant physique du nœuds (device): EUI-64 unique mondialement, attribué à la fabrication
  - DevAddr - 32 bit adresse du nœud: dynamique, non unique, attribuée grâce à la procédure d'activation (détails prochain cours)
  - GwEUI - 64 bit identifiant physique de la passerelle: EUI-64 (unique)
  - AppEUI- 64 bits identifiant de l'application: EUI-64, unique par réseau

- **Clés à connaître (détails en cours suivant):**
  - AppKey: clé primaire utilisée pour dériver les clés de sessions
  - AppSKey: clé de session de confidentialité (chiffrement des données)
  - NwkSKey: clé de session d'intégrité des données.

LoRaWAN : déploiement des rés. LoRaWAN

- **Réseaux opérées**
  - Passerelles, serveurs réseaux et serveurs applicatifs gérées par les opérateurs. Ex. en France: Orange et Objenious de Bouygues
- **Réseaux privées**
  - Plusieurs solutions de serveurs réseaux et applicatifs disponibles en open source
- **Réseaux communautaire**
  - Exemple du TTN (déploiement communautaire des passerelle), serveurs réseau et serveur applicatif gérés par TTN.

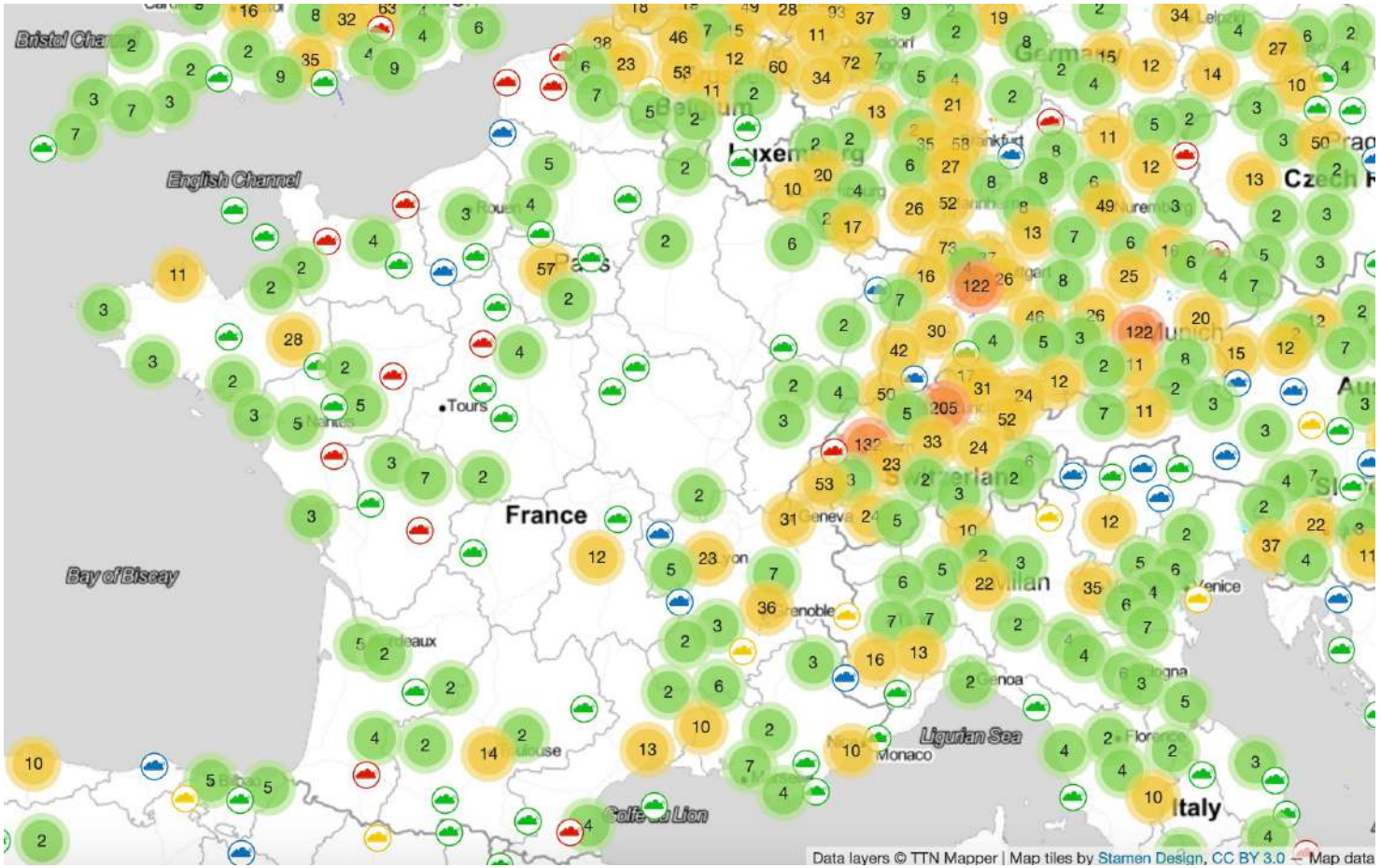
LoRaWAN : Couvertures



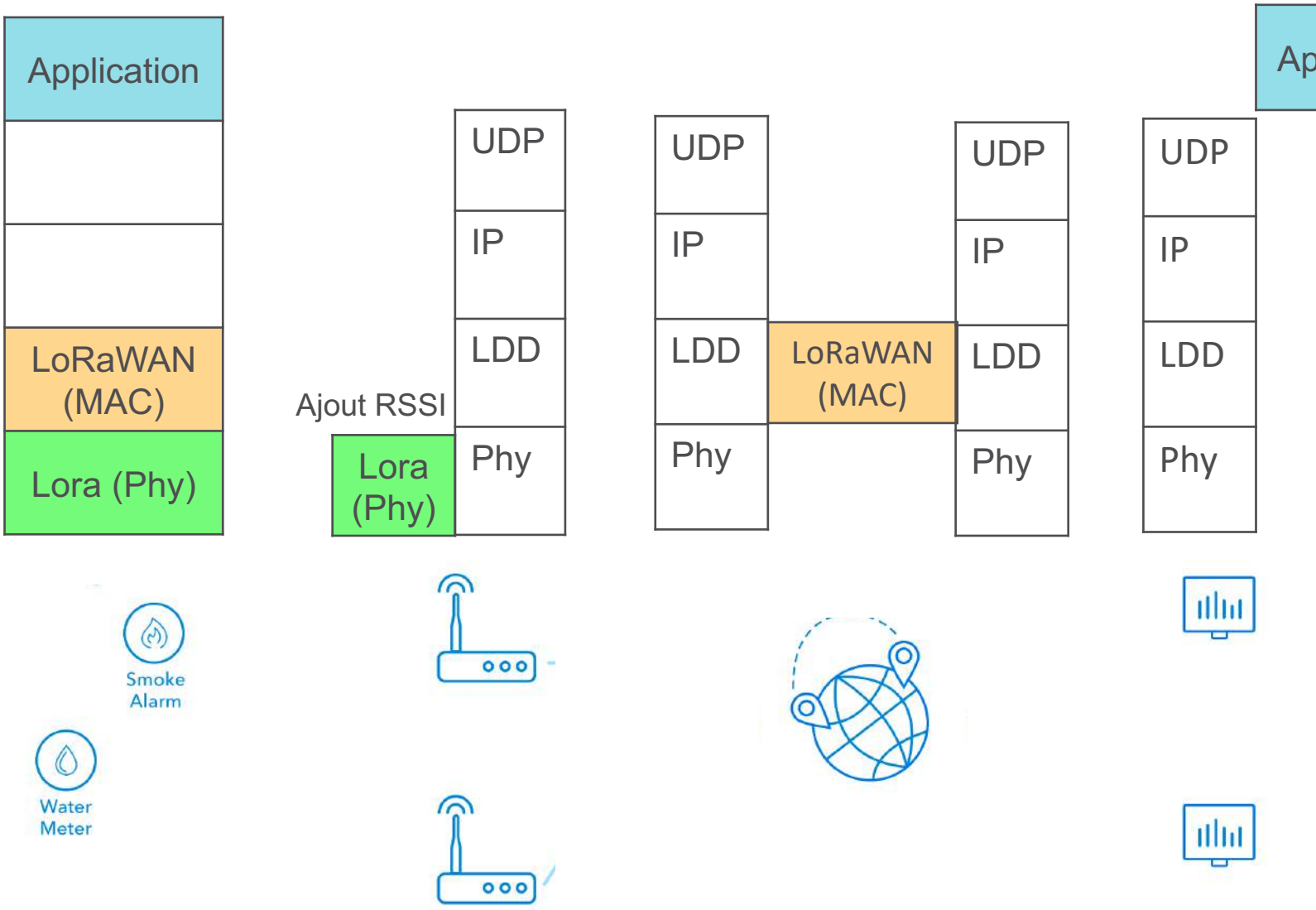
LoRaWAN : Couvertures



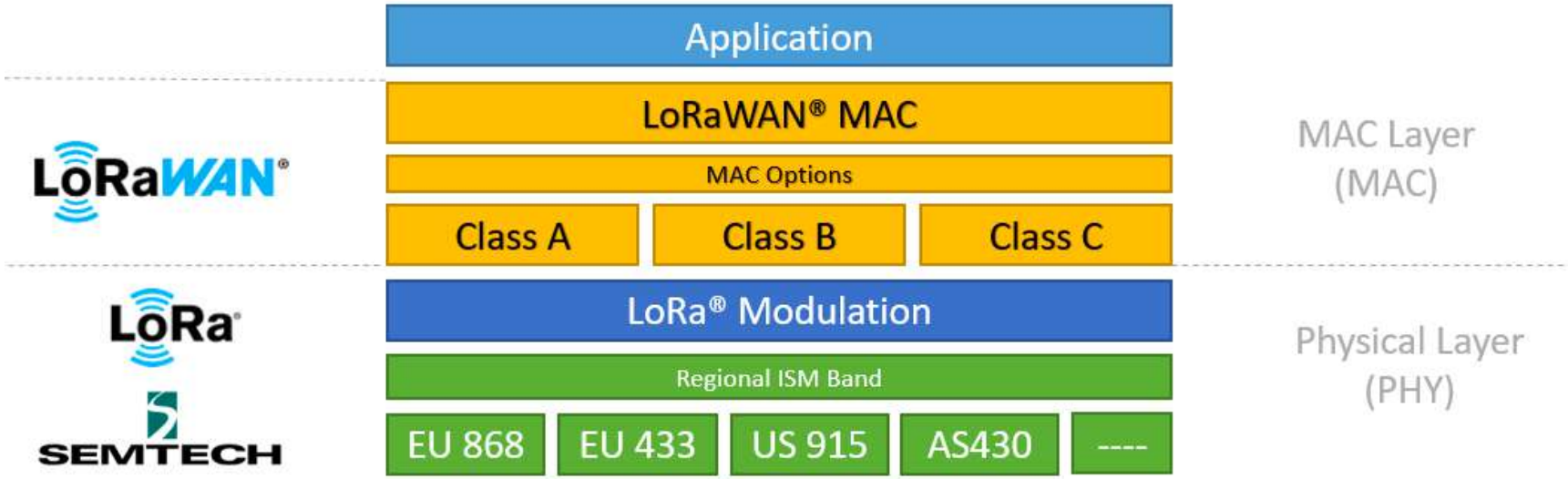
LoRaWAN : Couvertures



LoRaWAN : Architecture et protocole



LoRa/LoRaWAN



source: https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

LoRaWAN : Data Rate

- 8 Data Rate normalisés

DataRate	Configuration	Indicative physical bit rate [bit/s]
0	LoRa: SF12 / 125 kHz	250
1	LoRa: SF11 / 125 kHz	440
2	LoRa: SF10 / 125 kHz	980
3	LoRa: SF9 / 125 kHz	1760
4	LoRa: SF8 / 125 kHz	3125
5	LoRa: SF7 / 125 kHz	5470
6	LoRa: SF7 / 250 kHz	11000
7	FSK: 50 kbps	50000
8..15	RFU	

Source: **LoRaWAN 1.0.3 Regional Parameters, LoRa Alliance, 2018**

LoRaWAN : Plan de fréquences

EU863-870

Uplink:

1. **868.1** - SF7BW125 to SF12BW125
2. **868.3** - SF7BW125 to SF12BW125 and SF7BW250
3. **868.5** - SF7BW125 to SF12BW125
4. **867.1** - SF7BW125 to SF12BW125
5. **867.3** - SF7BW125 to SF12BW125
6. **867.5** - SF7BW125 to SF12BW125
7. **867.7** - SF7BW125 to SF12BW125
8. **867.9** - SF7BW125 to SF12BW125
9. **868.8** - FSK

Downlink:

- Uplink channels 1-9 (RX1)
- **869.525** - SF9BW125 (RX2)

Source: TTN selon LoRaWAN regional parameters document

LoRaWAN : Protocole – Accès au canal

- Pas de communications device to device !
- Seul communication device → gateway et gateway → device
- Pas d'attachement à une passerelle ni d'écoute

- Accès basé sur un mécanisme aléatoire (**Aloha**)

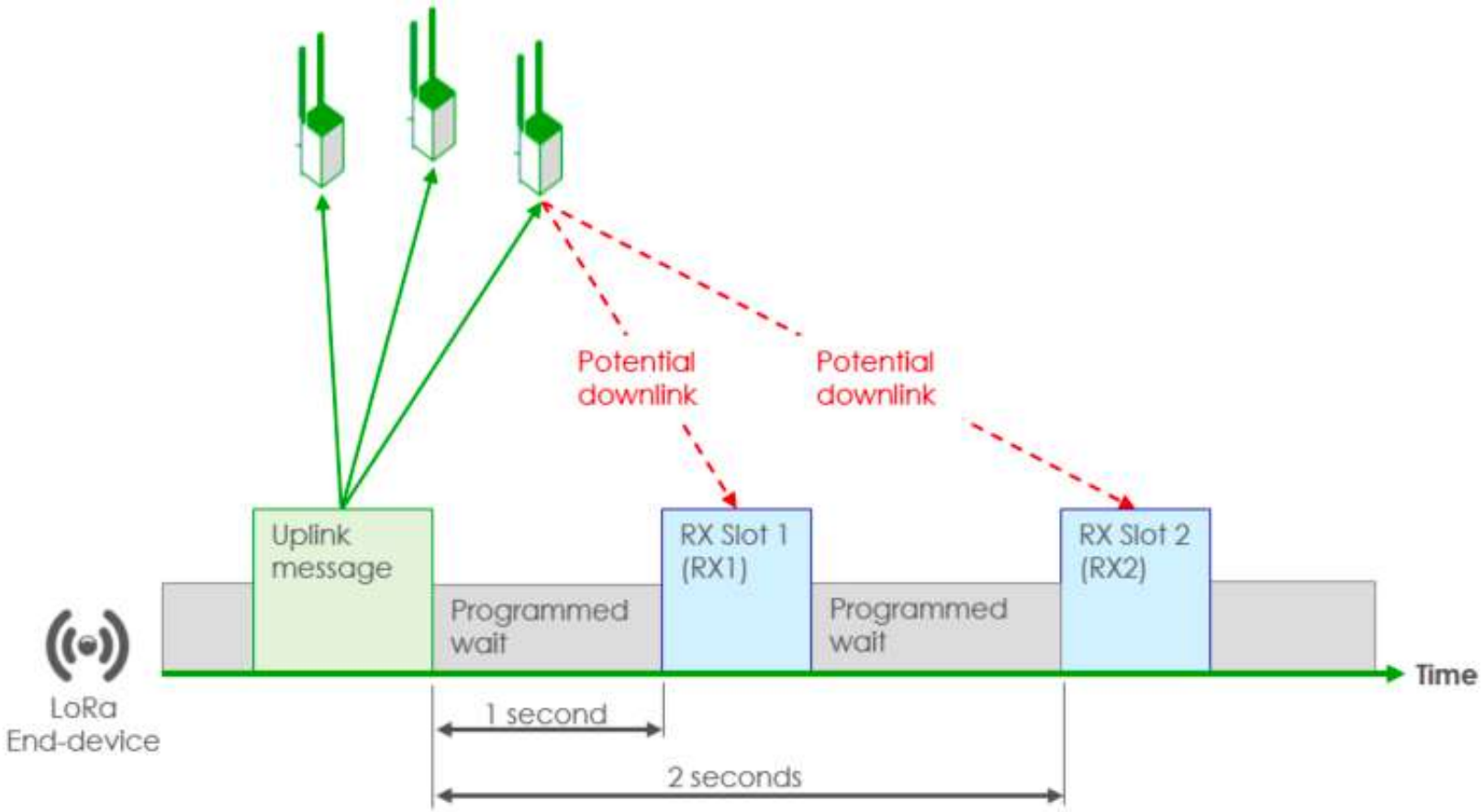
- Rappel: On peut décoder deux signaux envoyé au même temps avec deux SF différents ne sot pas en collision
- Un Canal logique = canal physique + Spreading Factor

LoRaWAN : ADR

- **ADR: Adaptative Data Rate**
- A la demande des nœuds (devices)
- Permet d'adapter le DR des nœuds pour proposer le débit le plus élevé possible au nœud
- Le serveur réseau calcule ce débit optimal sur la base de plusieurs mesures (20 dernière mesures sur TTN)
- Débit le plus rapide --> moins de temps de transmission → économie d'énergie
- Attention : n'est recommandé que lorsque les conditions radio sont très stables (souvent nœuds fixes), sinon ce mode est désactivé

LoRaWAN : Quid au Downlink ?

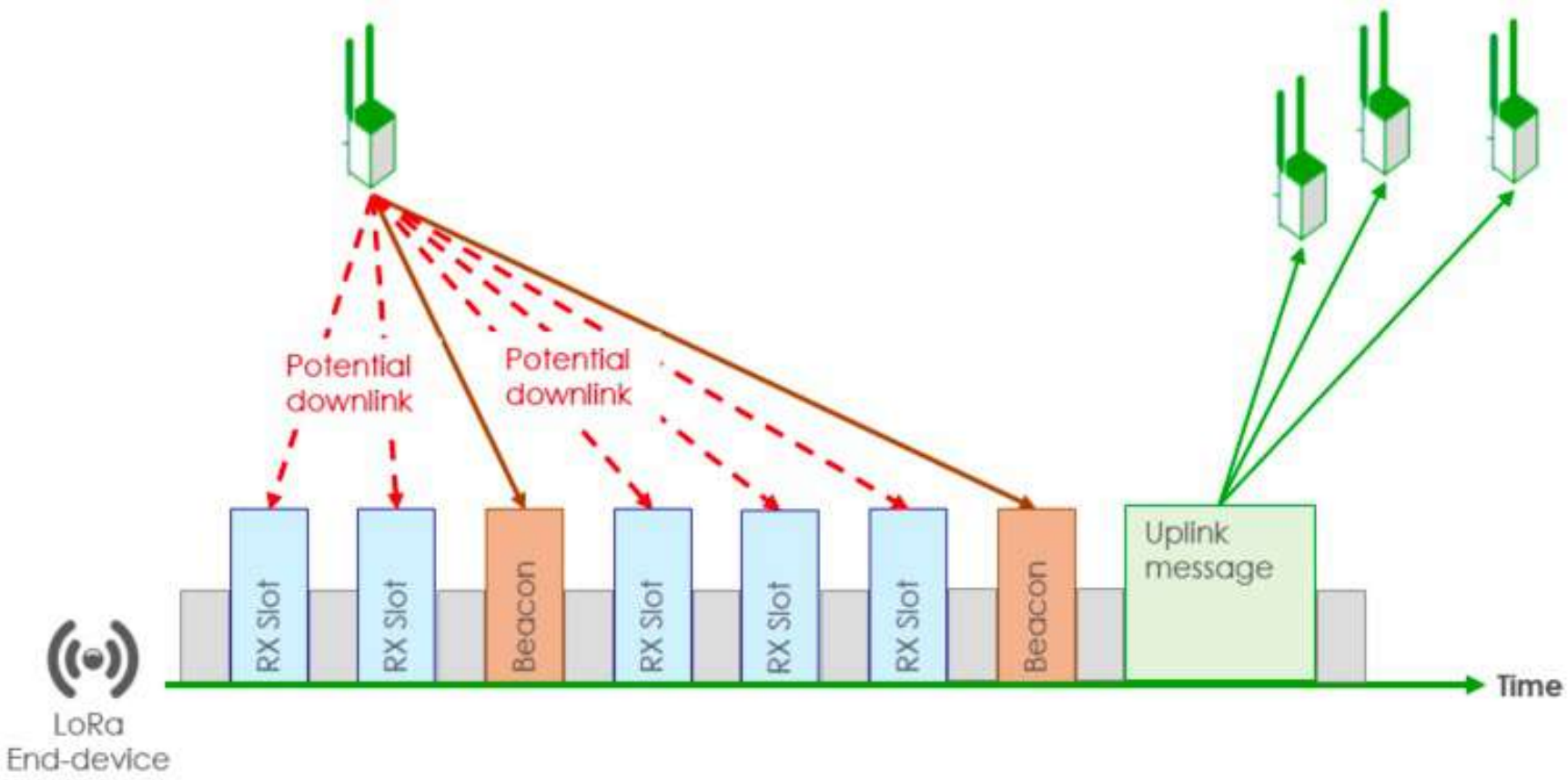
- **Trois classes de nœuds : Classe A**



source de l'image: https://zakelijkforum.kpn.com/lora-forum-16/lorawan-device-classes-a-b-and-c-10972

LoRaWAN : Quid au Downlink ?

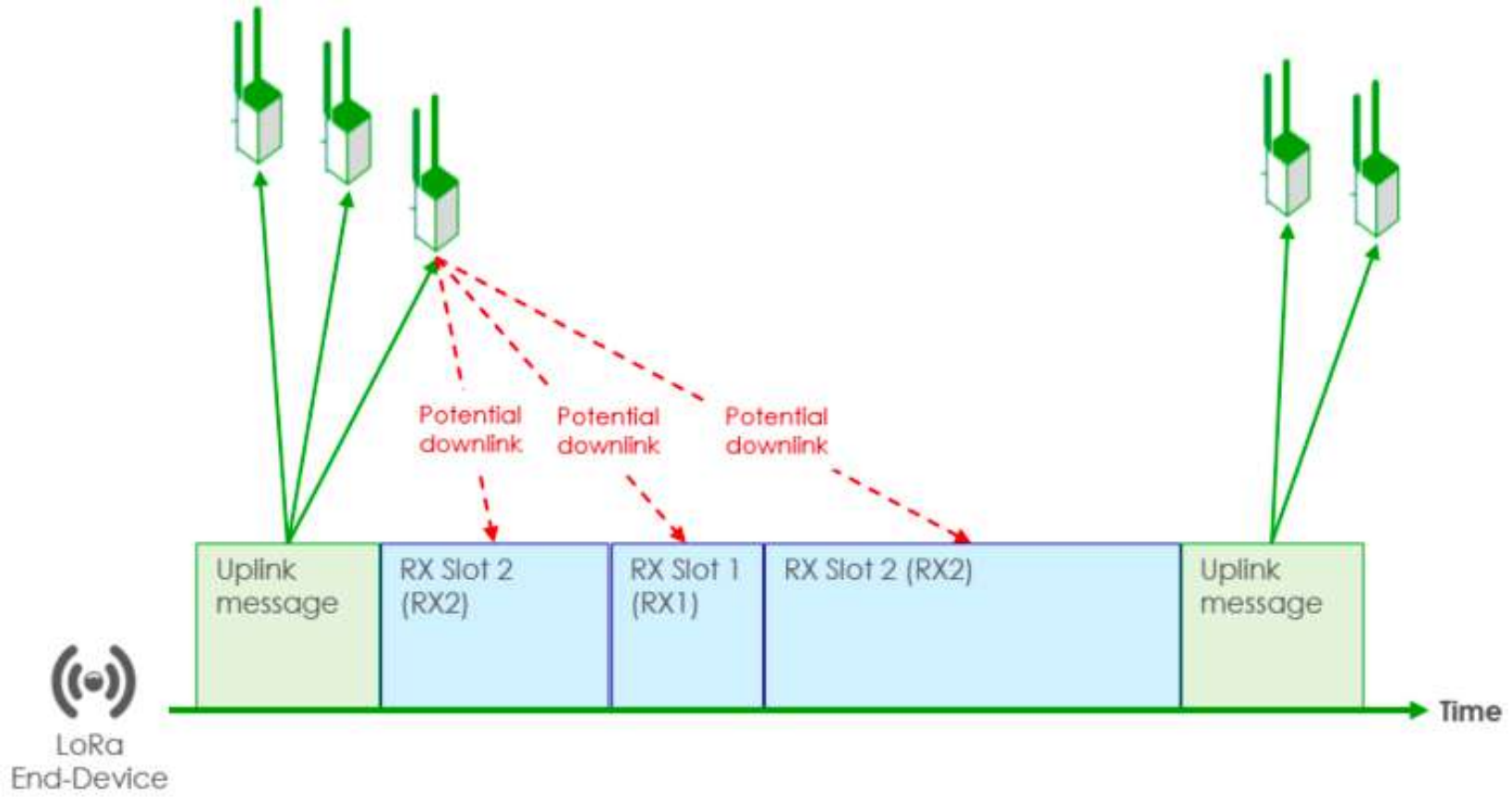
- **Trois classes de nœuds : Classe B**



source de l'image: https://zakelijkforum.kpn.com/lora-forum-16/lorawan-device-classes-a-b-and-c-10972

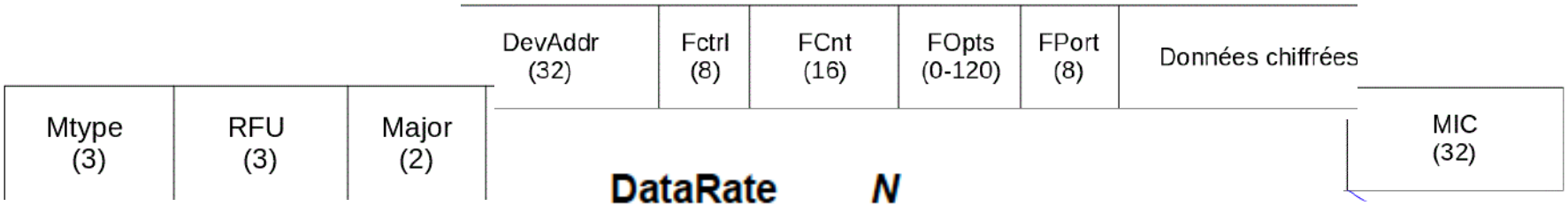
LoRaWAN : Quid au Downlink ?

- **Trois classes de nœuds : Classe C**



source de l'image: https://zakelijkforum.kpn.com/lora-forum-16/lorawan-device-classes-a-b-and-c-10972

LoRaWAN : Format de paquet et taille maximale



DataRate	*N*
0	51
1	51
2	51
3	115
4	222
5	222
6	222
7	222

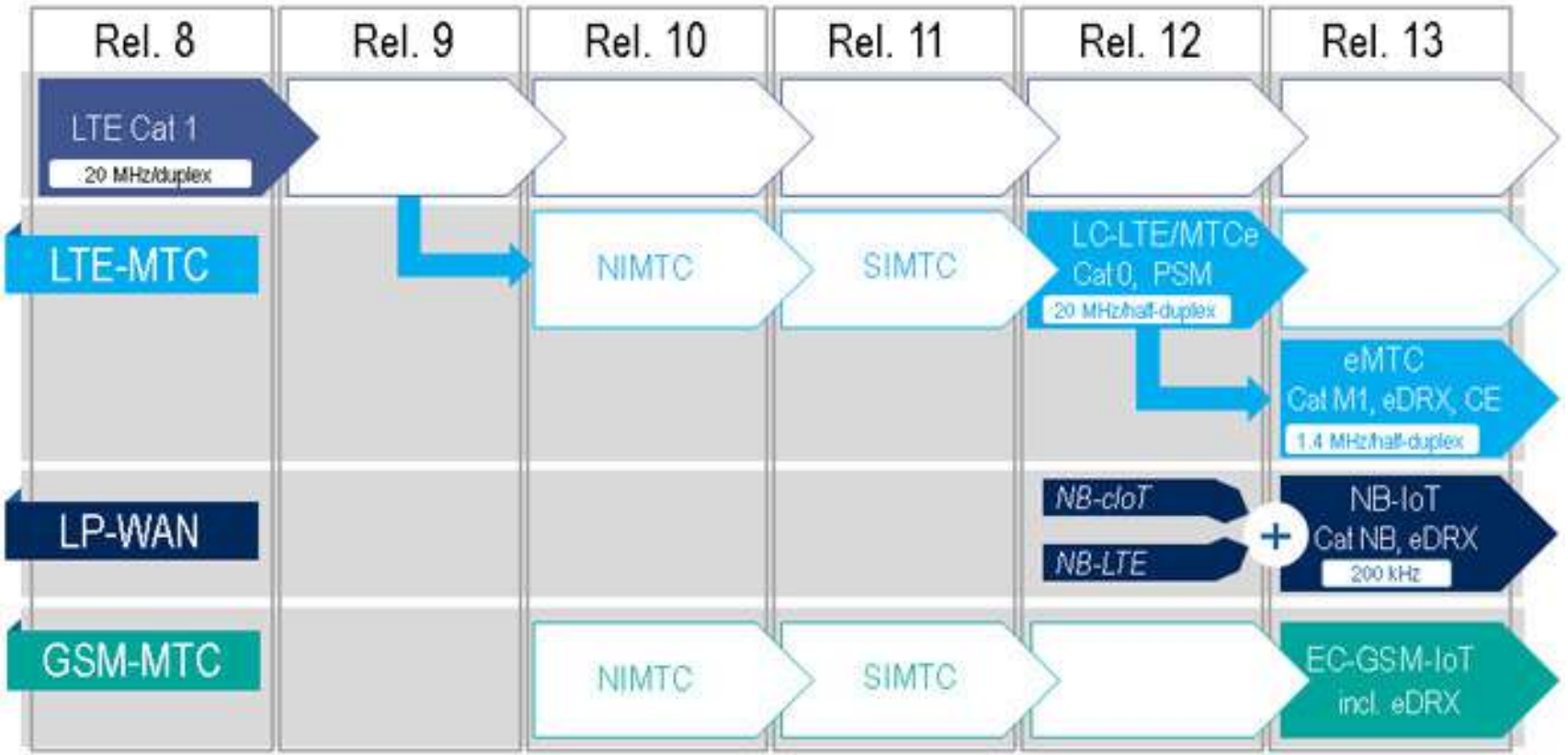
Source: **LoRaWAN 1.0.3 Regional Parameters, LoRa Alliance, 2018**

⚠ Attention: en plus des contraintes liées aux duty-cycle et aux taille de messages, l'opérateur peut imposer des contraintes plus fortes.

Exemple limites TTN:

Uplink : 30 sec / jour / nœud Downlink: 10 messages / jour / nœud

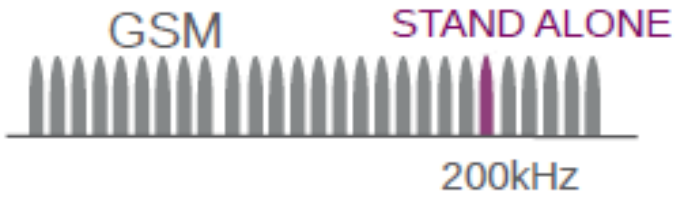
Fréquences sous licences: 3GPP IoT



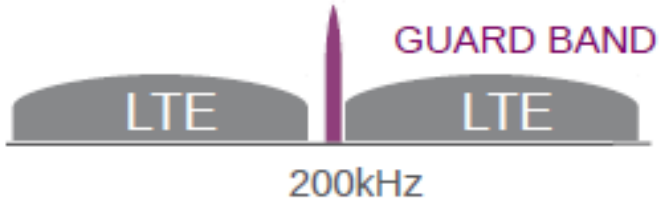
NB-IOT : Main actors: Intel, Nokia, Ericsson, Huawei, Gemalto, etc.

Fréquences sous licences: 3GPP IoT

Standard (release 13)	eMTC(LTE-M)	NB-IOT	EC-GSM-LTE
Deployment	InBand LTE	**In-band LTE, guard-band LTE, stanalone**	In-band GSM
Rate	1Mbps UL/DL	**250 kbps DL 20/250 kbps UL**	4 slots 70 kbps (GMSK) 240 (8PSK)
Bandwith	1,08 Mhz	**200 Khz**	200Khz

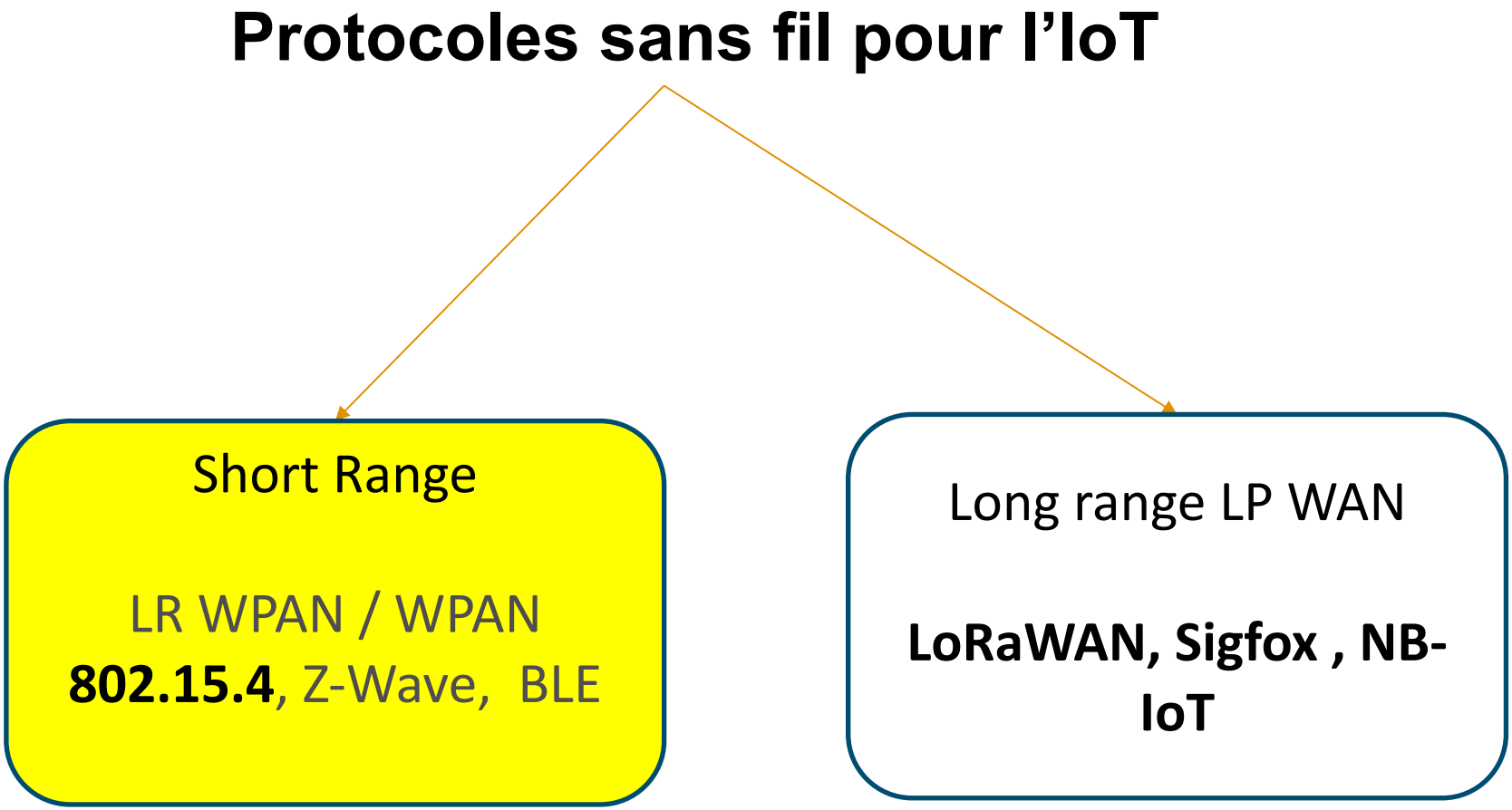




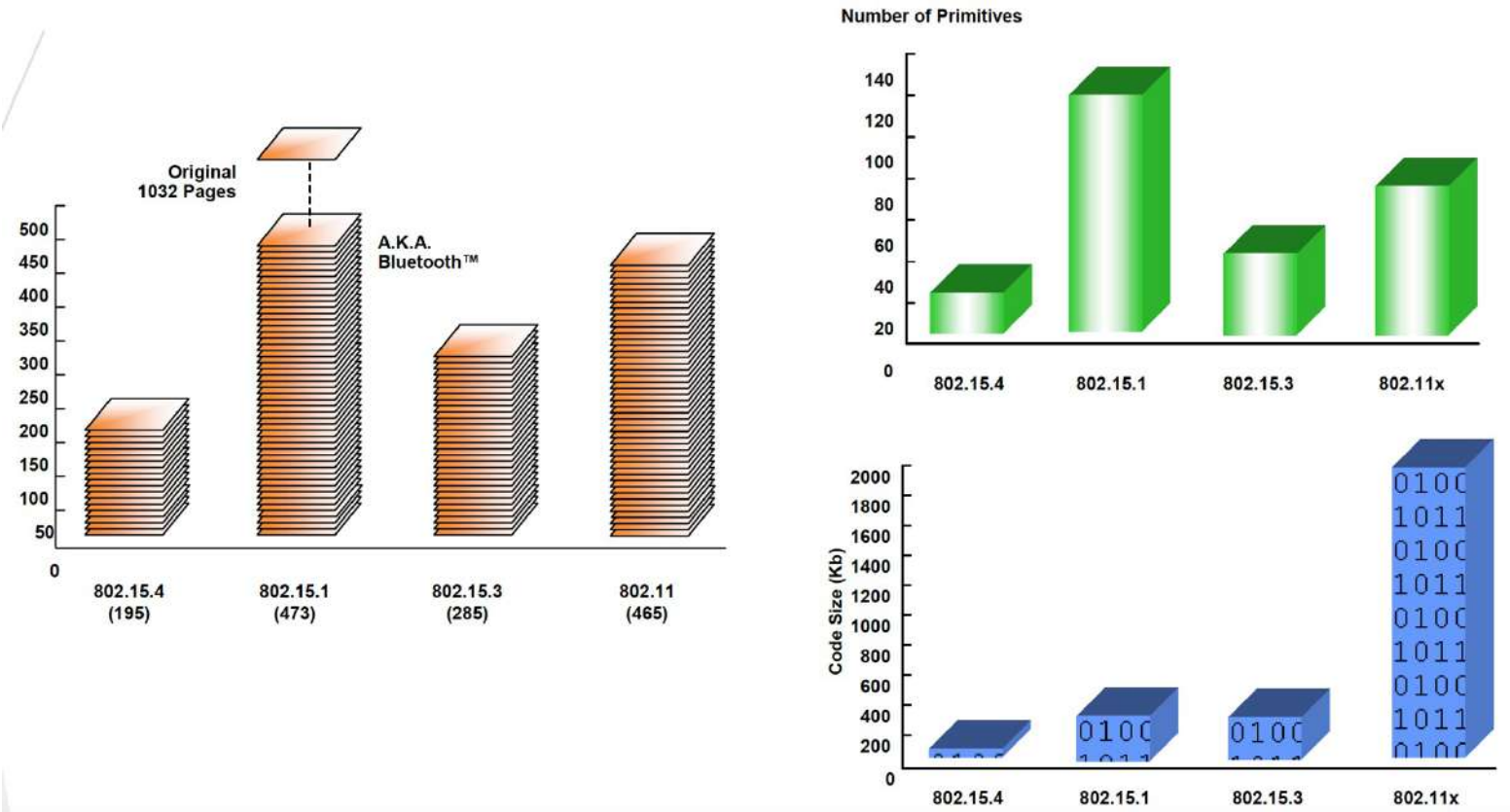


1.4 Solutions short range

Classification des protocoles IoT



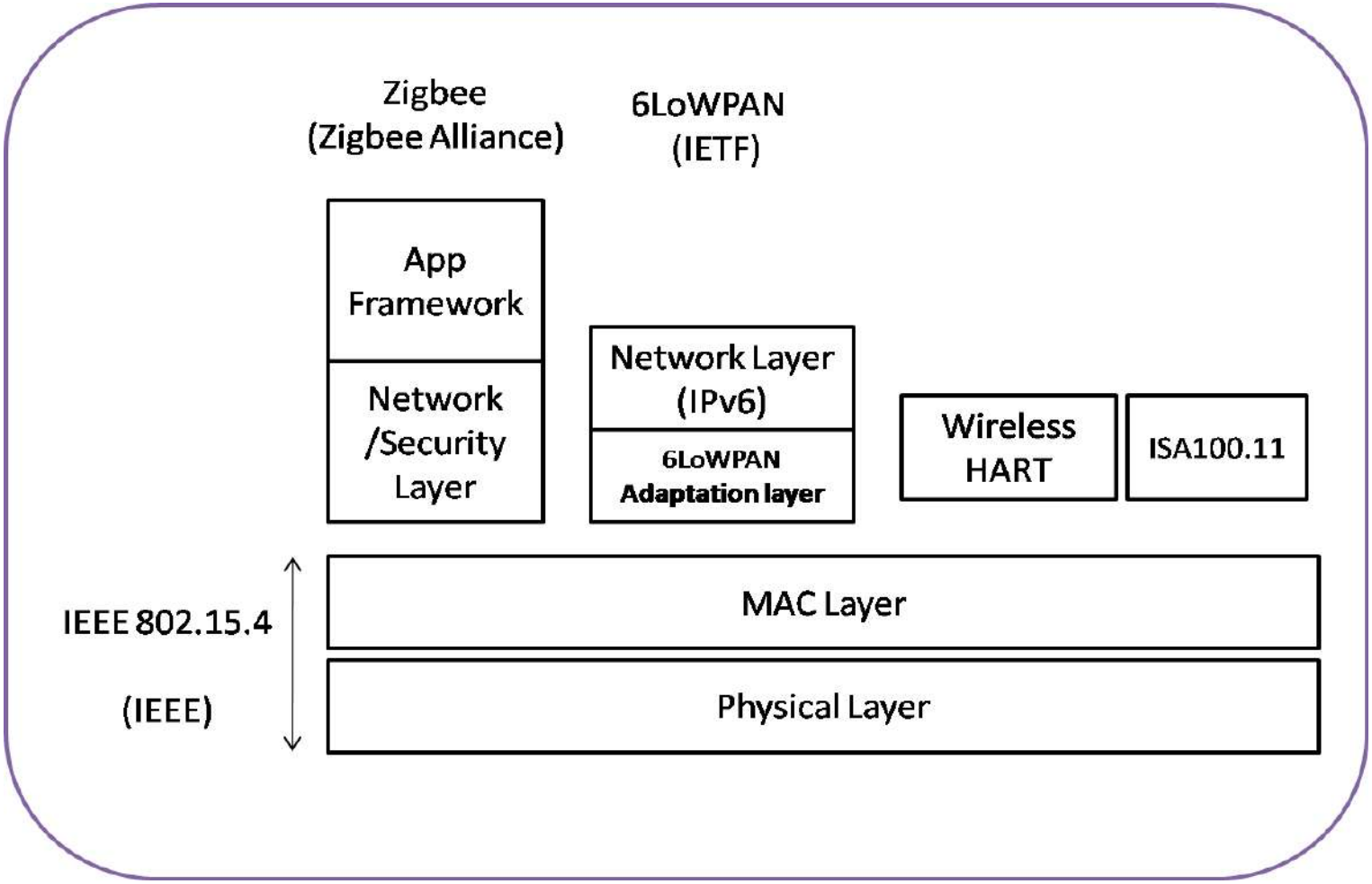
Classification des principaux standards



Source de la figure: José A. Gutierrez, IEEE Std. 802.15.4 Enabling pervasive Wireless Sensor Networks, 2015

LR WPAN: IEEE 802.15.4 et son positionnement

Exemple: IEEE 802.15.4



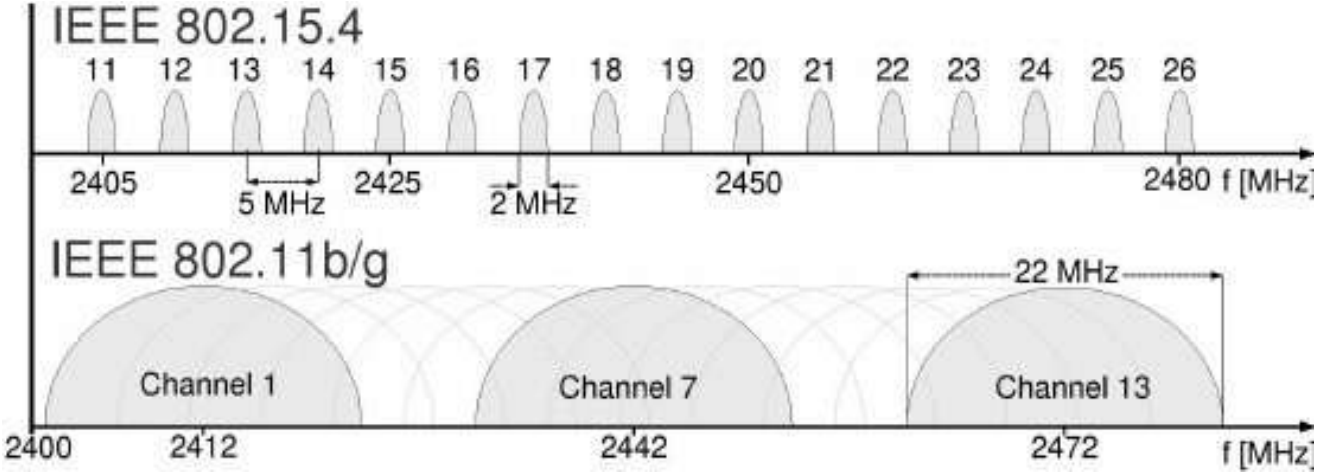
IEEE 802.15.4 Couche Physique

- **Couche physique** à faible complexité, consommation et à faible coût (mais à faible débit ☺)
- **Bandes sans licence :**

  868-868,6 MHz en Europe (1 canal, 20Kbps)

  902-928 MHz en Amérique (10 canaux, 40 kbps)

  2400-2483,5 MHz dans le monde (16 canaux, 250 kbp)



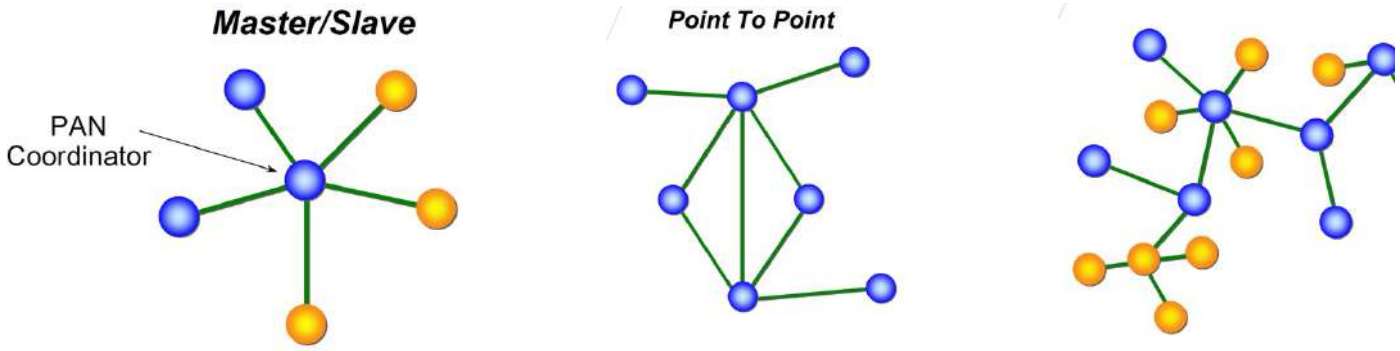
Source de la figure: Hauer, Jan-Hinrich & Handziski, Vlado & Wolisz, Adam. (2009). Experimental Study of the Impact of WLAN Interference on IEEE 802.15.4 Body Area Networks. 17-32.

802.15.4 Couche MAC (types et topologies)

- Deux classes de devices :

  FFD (full function device) : peut coordonner et peut communiquer avec tout autre nœud

  RFD (Reduced function device): simple implémentation, ne peut communiquer qu'avec un FFD.



Trois types de nœuds : Coordinateur du PAN (FFD), Coordinateur (FFD), et network device (RFD ou FFD)

Source de la figure: José A. Gutierrez, IEEE Std. 802.15.4 Enabling pervasive Wireless Sensor Networks, 2015

802.15.4 Couche MAC (identifiants et Trame)

- **Adresses EUI-64:** Chaque dispositif qui supporte 802.15.4 doit avoir une adresse EUI-64 unique au monde (équivalent adresse MAC)
- Les dispositifs sont regroupés dans des PAN identifiés sur deux octets
- Chaque dispositif peut être configuré avec une adresse courte sur 2 octets

802.15.4 Couche MAC (identifiants et Trame)

Bits: 0–2	3	4	5	6	7–9	10–11	12–13	14–15
Frame Type	Security Enabled	Frame Pending	AR	PAN ID Compression	Reserved	Dest. Addressing Mode	Frame Version	Source Addressing Mode

Octets: 2	1	0/2	0/2/8	0/2	0/2/8	0/5/6/10/14	variable	2
Frame Control	Sequence Number	Destination PAN Identifier	Destination Address	Source PAN Identifier	Source Address	Auxiliary Security Header	Frame Payload	FCS
		Addressing fields						
MHR							MAC Payload	MFR

127 octets max

Source de la figure : Lopez, Javier & Rios, Ruben & Bao, Feng & Wang, Guilin. (2017). Evolving privacy: From sensors to the Internet of Things. Future Generation Computer Systems. 75.

More details : https://core.ac.uk/download/pdf/14699804.pdf and https://scdn.rohde-schwarz.com/ur/pws/dl_downloads/dl_application/application_notes/1gp105/1GP105_1E_Generation_of_IEEE_802154_Signals.pdf

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		Addressing fields						
MHR							MAC Payload	MFR

Champs d'adressage

127 octets max

Source de la figure : Lopez, Javier & Rios, Ruben & Bao, Feng & Wang, Guilin. (2017). Evolving privacy: From sensors to the Internet of Things. Future Generation Computer Systems. 75.

More details : https://core.ac.uk/download/pdf/14699804.pdf and https://scdn.rohde-schwarz.com/ur/pws/dl_downloads/dl_application/application_notes/1gp105/1GP105_1E_Generation_of_IEEE_802154_Signals.pdf

802.15.4 Couche MAC (identifiants et Trame)

Bits: 0–2	3	4	5	6	7–9	10–11	12–13	14–15
Frame Type	Security Enabled	Frame Pending	AR	PAN ID Compression	Reserved	Dest. Addressing Mode	Frame Version	Source Addressing Mode

Octets: 2	1	0/2	0/2/8	0/2	0/2/8	0/5/6/10/14	variable	2
Frame Control	Sequence Number	Destination PAN Identifier	Destination Address	Source PAN Identifier	Source Address	Auxiliary Security Header	Frame Payload	FCS
		Addressing fields						
MHR							MAC Payload	MFR

Champs sécurité

127 octets max

Source de la figure : Lopez, Javier & Rios, Ruben & Bao, Feng & Wang, Guilin. (2017). Evolving privacy: From sensors to the Internet of Things. Future Generation Computer Systems. 75.

More details : https://core.ac.uk/download/pdf/14699804.pdf and https://scdn.rohde-schwarz.com/ur/pws/dl_downloads/dl_application/application_notes/1gp105/1GP105_1E_Generation_of_IEEE_802154_Signals.pdf

802.15.4 Couche MAC: Accès au canal

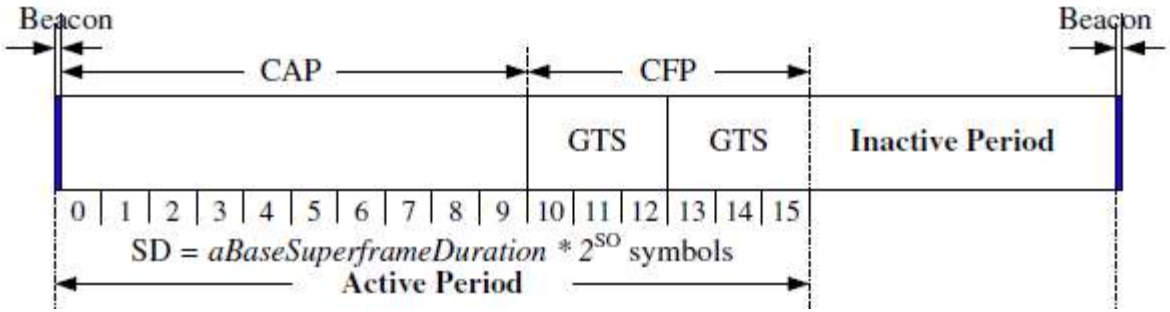
- **Non-beacon-enabled mode :** CSMA-CA non “slottée”
- **Beacon-enabled mode :** CSMA-CA “slottée”

  Une super-trame envoyée par coordinateur du PAN:

  – A Contention Access Period (CAP): CSMA/CA slottée;

  – A Contention-Free Period (CFP): Slots à accés guaranti (GTS)

  – Une période d’inactivité optionnelle



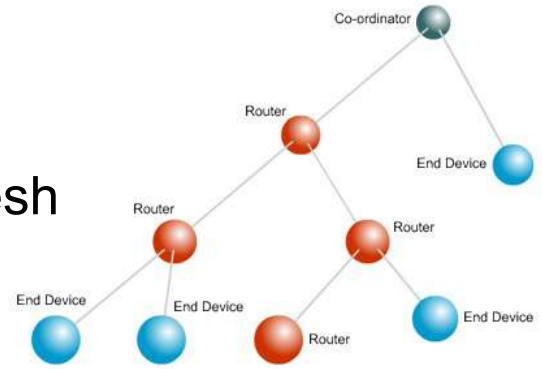
aBaseSuperframeDuration = 15.36 ms (for 250 kbps)
0 ≤ SFO ≤14

1.3 Solutions short range

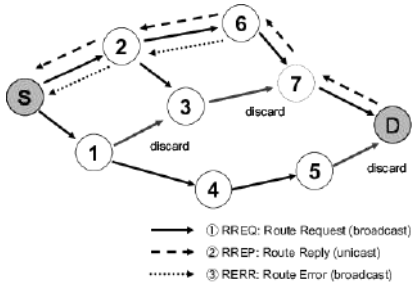
IEEE 802.15.4 et protocoles de couches supérieures

Zigbee

- **Trois types de noeuds**
  - –- ZigBee coordinator (ZC): Initialisation maintien et contrôle du réseau
  - – ZigBee router (ZR): routage des paquets
  - – ZigBee end device (ZED) : capteurs d'extrimité
- **Trois types de topologies :** star, tree and mesh



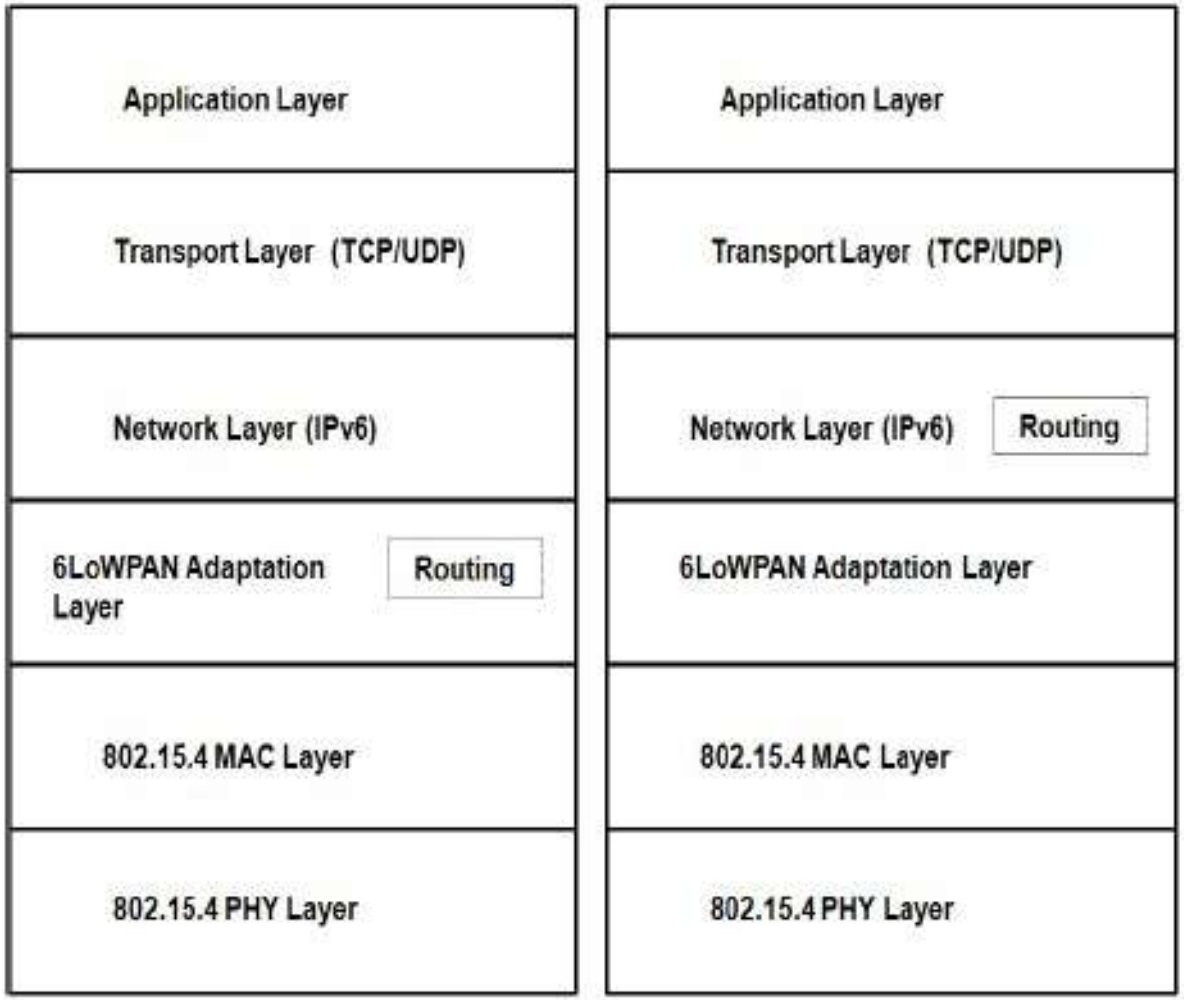
- **Topologie en arbre:** routage basé sur les adresses attribuées de telle sorte à pouvoir déduire le parent automatiquement
- **Topologie en mesh:** routage à travers un protocole AODV réactif



6LowPAN

- Adaptation d'IPv6 pour prendre en charge les dispositifs à faible consommation d'énergie et à faible coût
- A implémenter au-dessus du standard IEEE 802.15.4
- Couche d'adaptation pour fragmenter les paquets IP en trames IEEE 802.15.4 (127 octets maximum)
- 6LowPan définit les compressions d'en-tête IPv6
- 6LowPAN : Interopérabilité Vs complexité et délai

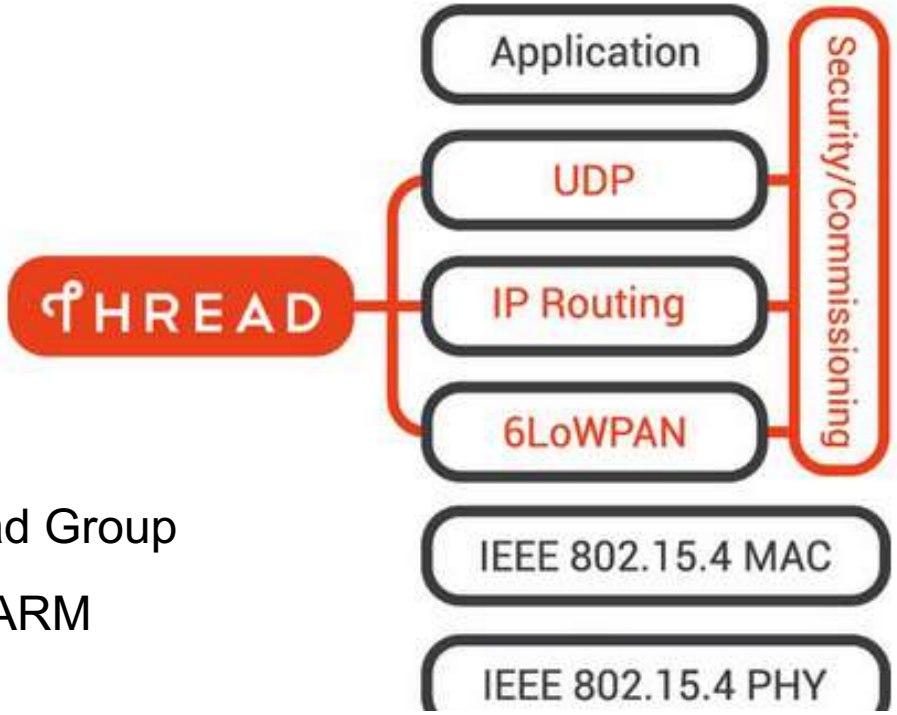
6LowPAN: Routage



6LoWPAN Mesh-under Routing

6LoWPAN Route-over Routing

6LowPAN & Thread

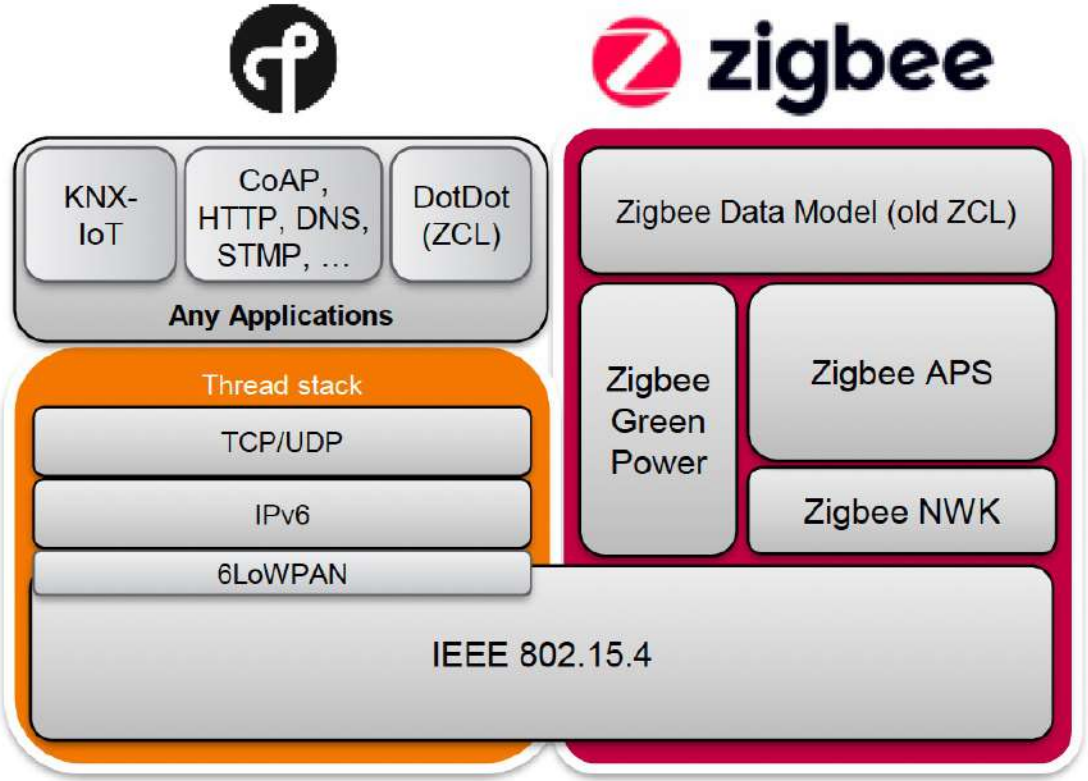


2014, lancement l'alliance Thread Group

Google (Nest Labs), Samsung, ARM Holdings, Qualcomm, etc.

2018: Apple rejoint le groupe ..

Thread Vs Zigbee



Introduction to 802.15.4-based solution for industrial IoT usage at Schneider Electric, Elodie Chargy , Nicolas Cochard, TC, 2021

Matter project

- **December 2019**, Amazon, Apple, Google, Samsung SmartThings, the Zigbee Alliance → Project Connected Home over IP (CHIP)
- **May 2021:** Zigbee Alliance → Connectivity Standards Alliance

  Project Connected Home over IP (CHIP) → **Matter**

Matter : IP based with and an open-source approach

30/09/2022 - Publication de la version 1.0 du standard Matter

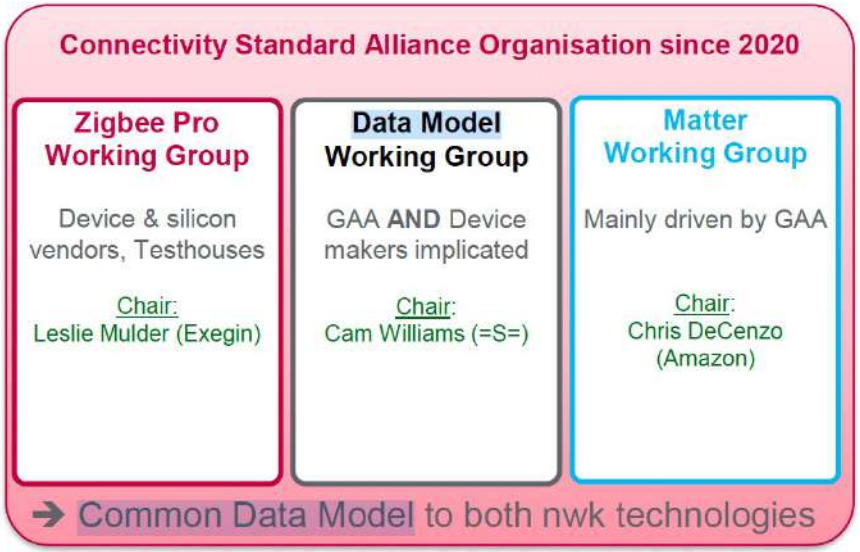
23/10/2023 - Publication de la version 1.2 du standard Matter



Source : https://buildwithmatter.com/ ☺

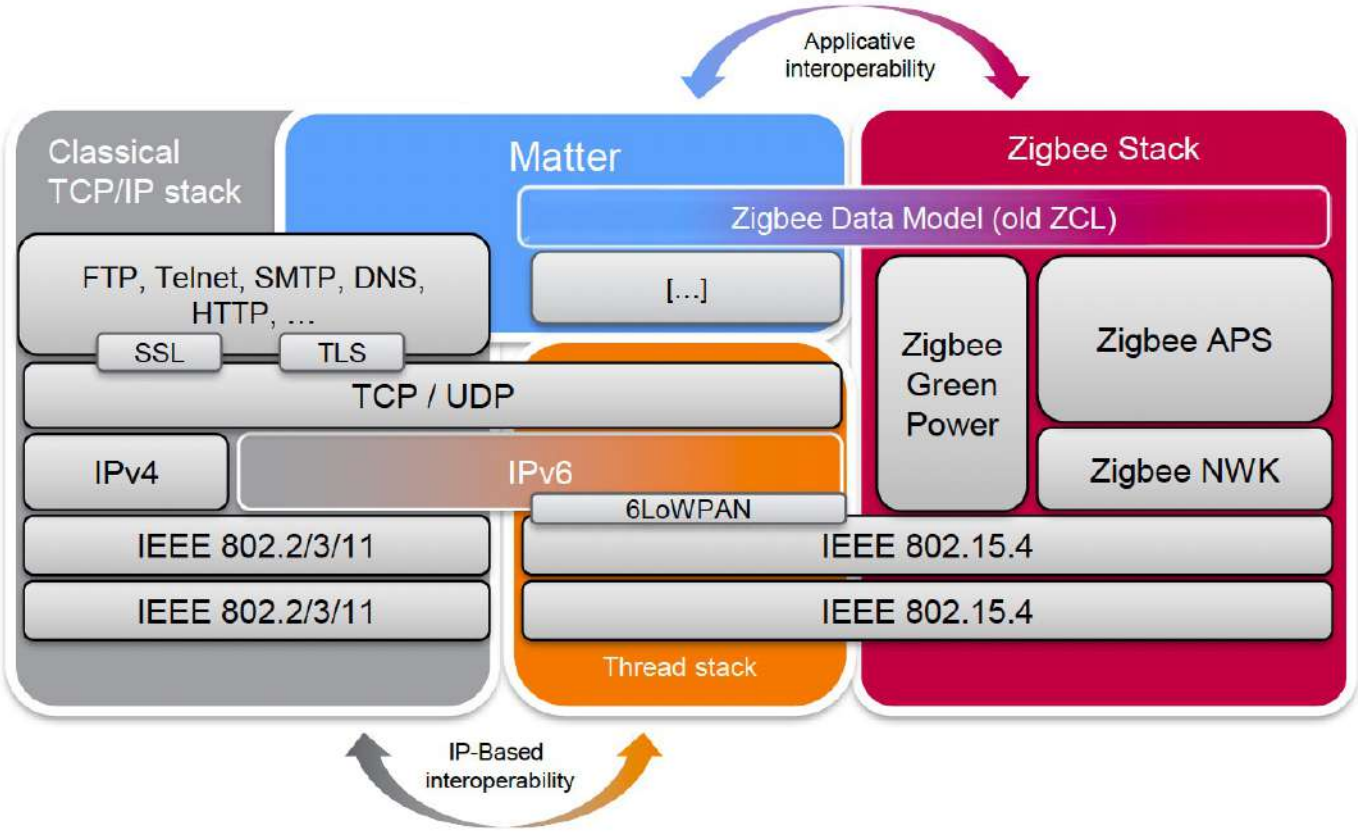
Positionnement du CSA





Introduction to 802.15.4-based solution for industrial IoT usage at Schneider Electric, Elodie Chargy , Nicolas Cochard, TC, 2021

Matter, Thread and Zigbee



Introduction to 802.15.4-based solution for industrial IoT usage at Schneider Electric, Elodie Chargy , Nicolas Cochard, TC, 2021

Principales références

- José A. Gutierrez, IEEE Std. 802.15.4 Enabling pervasive Wireless Sensor Networks, 2015.
- Hauer, Jan-Hinrich & Handziski, Vlado & Wolisz, Adam. (2009). Experimental Study of the Impact of WLAN Interference on IEEE 802.15.4 Body Area Networks. 17-32.
- Dedicated networks for IoT : PHY / MAC state of the art and challenges C. Goursaud, JM. Gorce, EAI Endorsed Transactions on the Internet of Things, 2016
- Lopez, Javier & Rios, Ruben & Bao, Feng & Wang, Guilin. (2017). Evolving privacy: From sensors to the Internet of Things. Future Generation Computer Systems. 75.
- Cours Fomation LoRa / LoRaWAN, Sylvain MONTAGNY, Université Savoie Mont Blanc
- Caractérisation des procédés de modulation et de démodulation LoRa, J-F SERREAU, 2019
- http://genelaix.free.fr/IMG/pdf/presentation_lora_lorawan.pdf
- http://www.onditech.com/documents/freq_433_868.pdf

Principales références

- LoRaWAN Specification v1.1 , LoRa Alliance, 2017
- LoRaWAN 1.0.3 Regional Parameters, LoRa Alliance, 2018
- Cours Fomation LoRa / LoRaWAN, Sylvain MONTAGNY, Université Savoie Mont Blanc
- Augustin, Aloÿs & Yi, Jiazi & Clausen, Thomas Heide & Townsley, William. (2016). A Study of LoRa: Long Range & Low Power Networks for the Internet of Things
- https://www.thethingsnetwork.org/
- http://genelaix.free.fr/IMG/pdf/presentation_lora_lorawan.pdf
- http://www.onditech.com/documents/freq_433_868.pdf
- https://zakelijkforum.kpn.com/lora-forum-16/lorawan-device-classes-a-b-and-c-10972
- https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

2- Sécurité de l'IOT

Security challenges in WSN

Information Security

$Risk = Threat \times Vulnerability \times Cost$

Vulnerability

- Deployment environment
- Lack of physical protection
- Memory limitation
- Energy limitation
- Wireless Communication
- Lack of infrastructure

Threat

- Traditional threats (eavesdropping, message injection ...)
- New specific attacks (energy drain, data aggregation attacks)

Cost

- Critical applications of WSN
- Dramatic effects and costs

Ex: : Mirai botnet (DDoS attacks in 2016/17)

Security challenges in WSN

IT life cycle

- Factorized software
- Upgrades
- Soft restes when unusable
- Large scale
- Mature codebase

IoT life cycle

- Fragmentation
- No upgrades
- Non soft restes
- Very large scale
- Young codebase

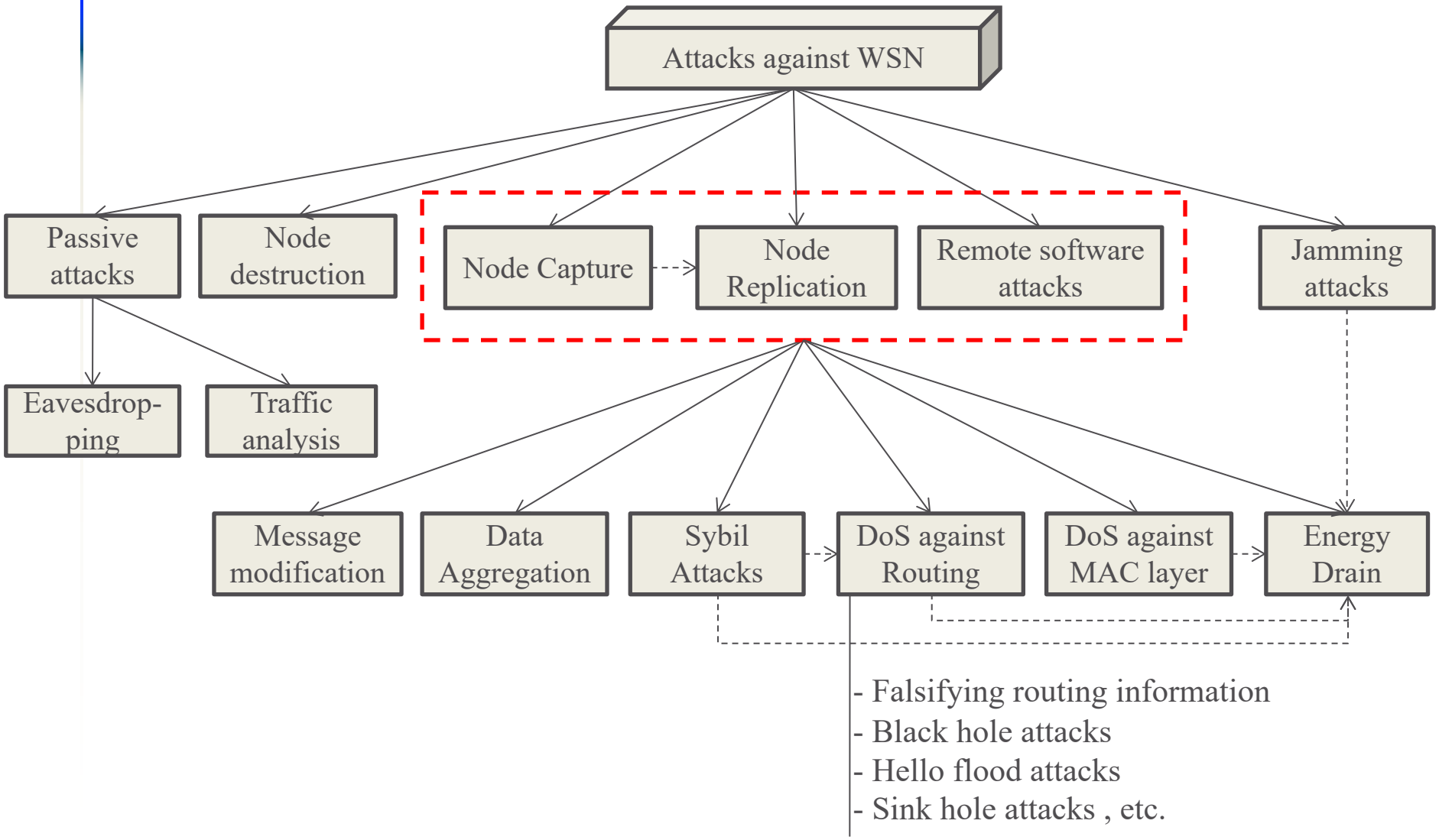
Source: How IoT revisit security problems, François Lesueur, INSA-Lyon, April 2017

It's not the walls that protect the citadel, but the spirit of its inhabitants."

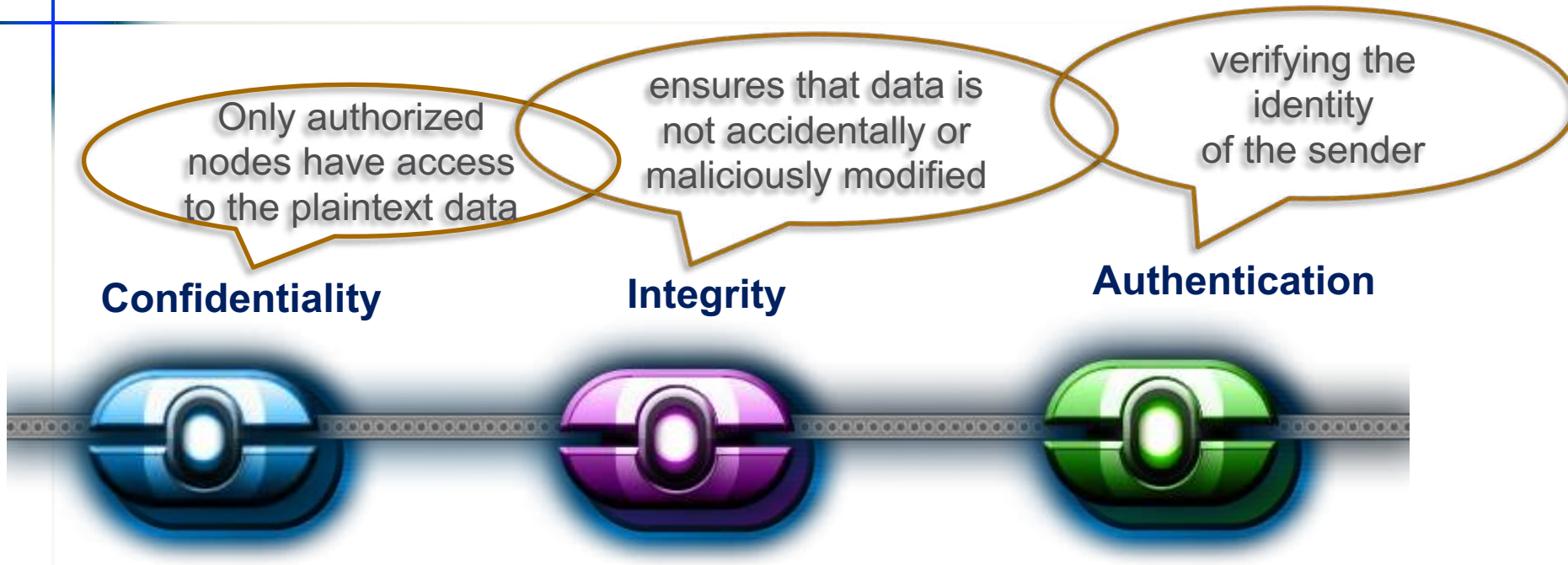
[Thucydides]

At least 40% of attacks are caused by users themselves.

Attacks against WSN



Security requirement in WSN



- **Basic services**
- **Securing routing**
- **Securing data aggregation**



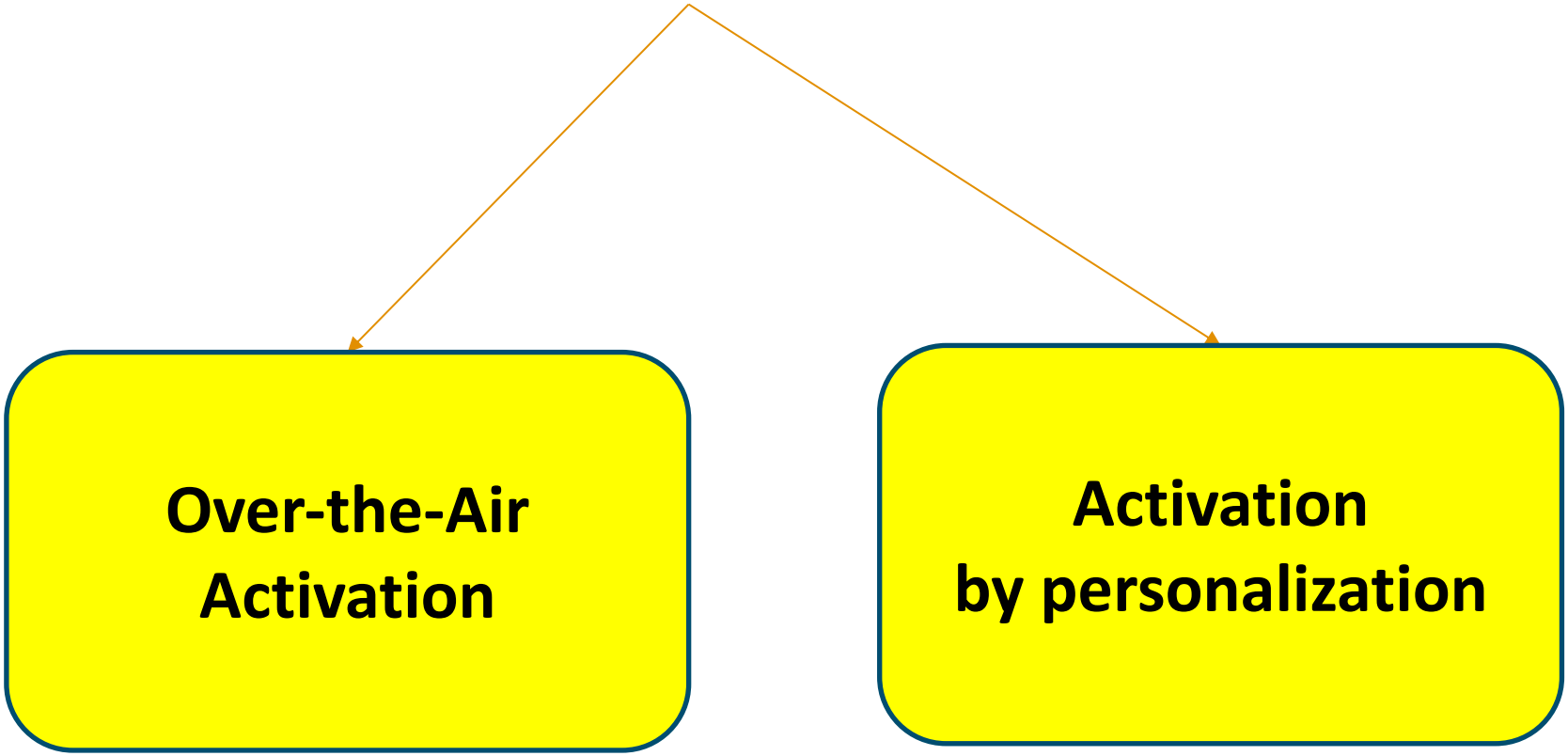
Key management:

A corner stone

2- Sécurité de l'IOT

2.1 Réseaux LoRaWAN

Sécurité LoRaWAN



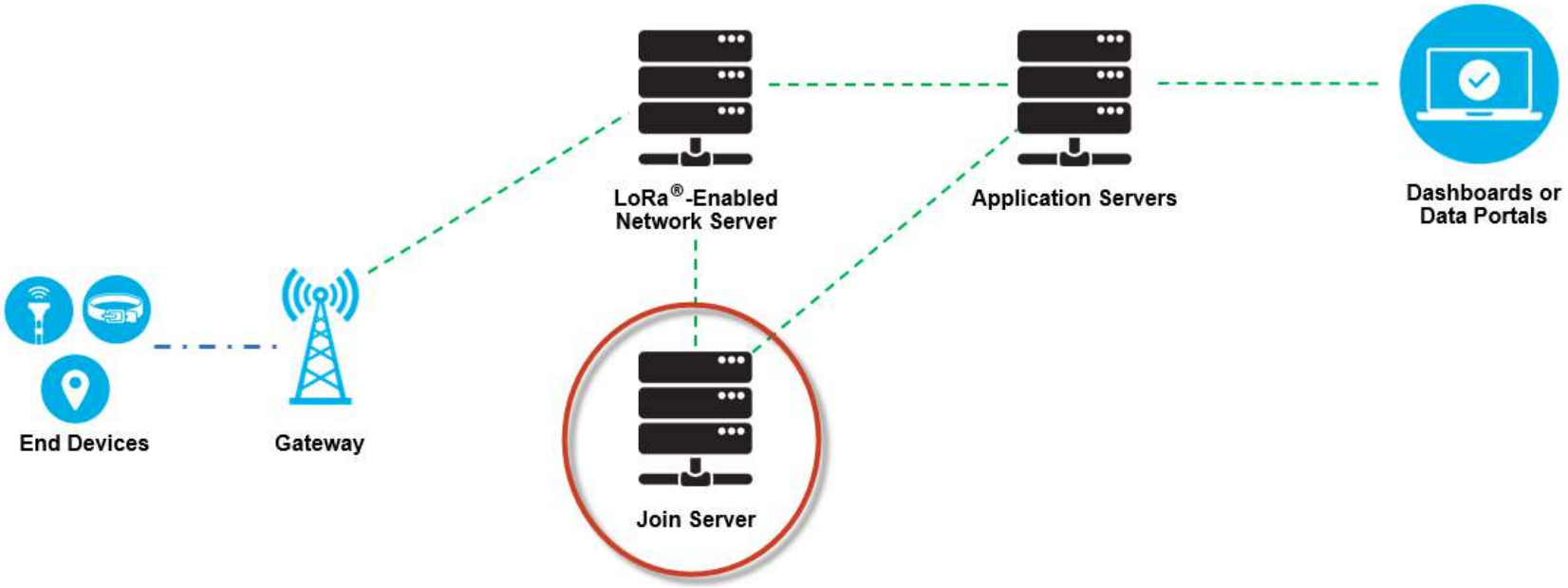
LoRaWAN Security

LoRaWAN security: Two modes

- **OTAA (Over-the-Air Activation)**
  - Master 128-bit Key (**Appkey**) for initial mutual authentication and the generation of two **sessions keys**:
    - Network Session Key (**NwkSKey**) for message integrity and authentication
    - Application Session Keys (**AppSKey**) for encryption (confidentiality)
- **ABP (Activation by personalization)**
  - each end-node is provided with **NwkSKey and AppSKey**

LoRaWAN Security: OTTA

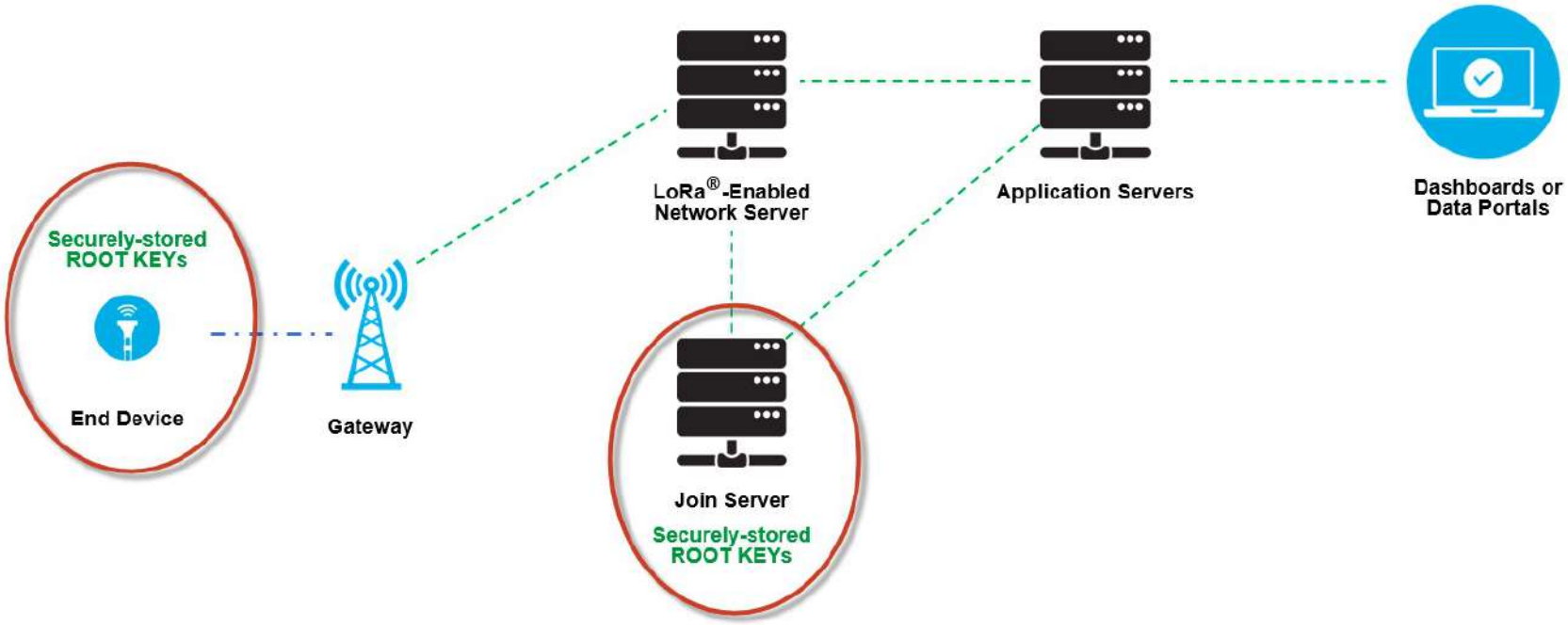
Join Server :



source: https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

LoRaWAN Security: OTTA

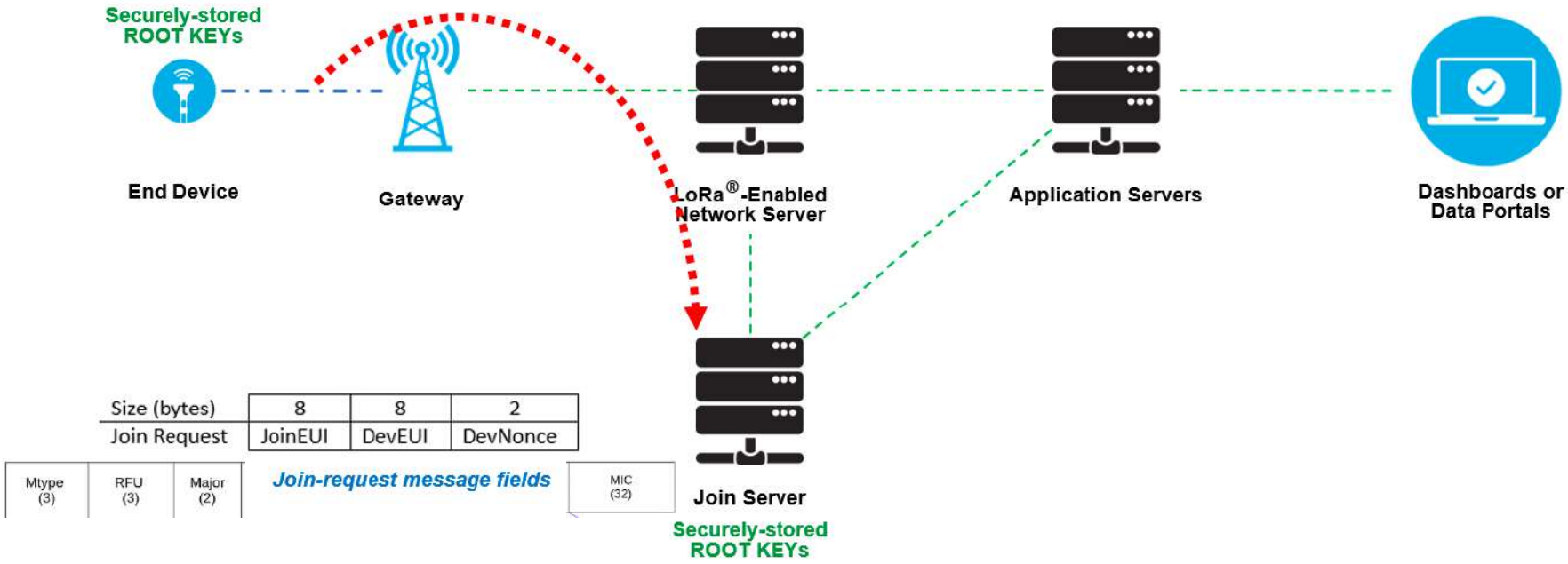
Root key: AppKey



source: https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

LoRaWAN Security: OTTA

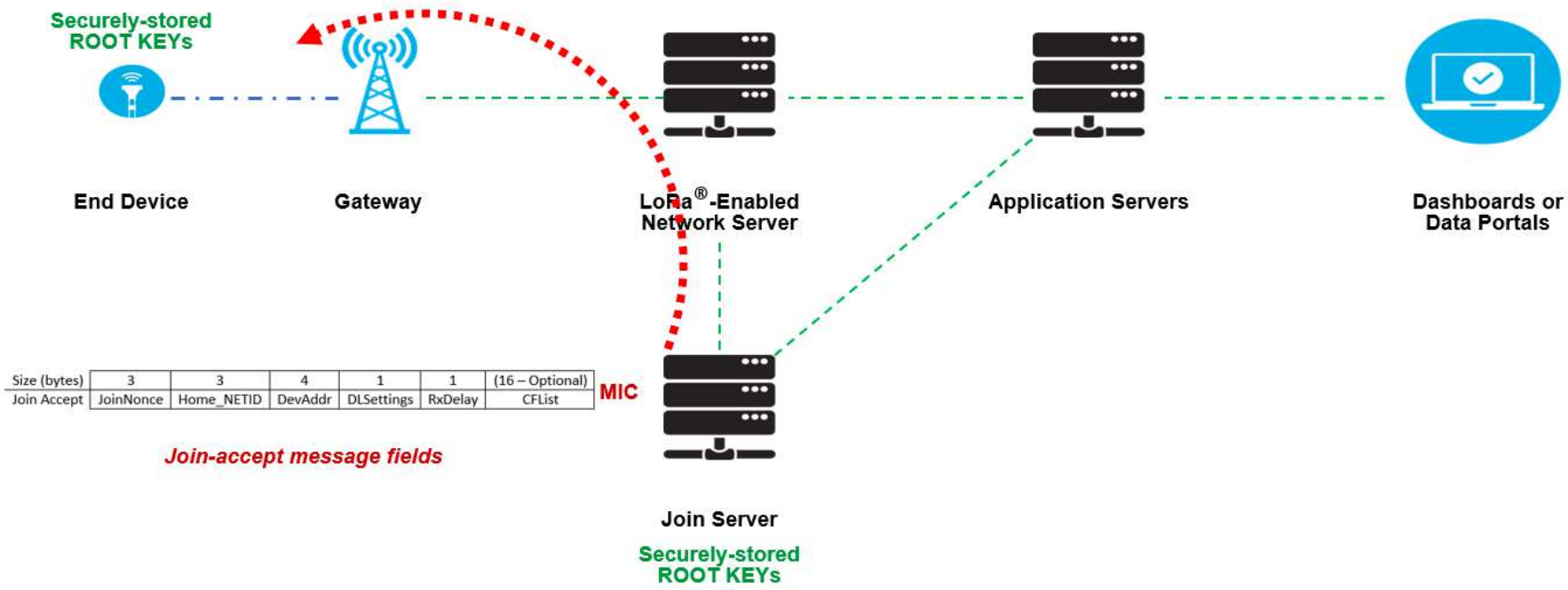
Join Request



source: https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

LoRaWAN Security: OTTA

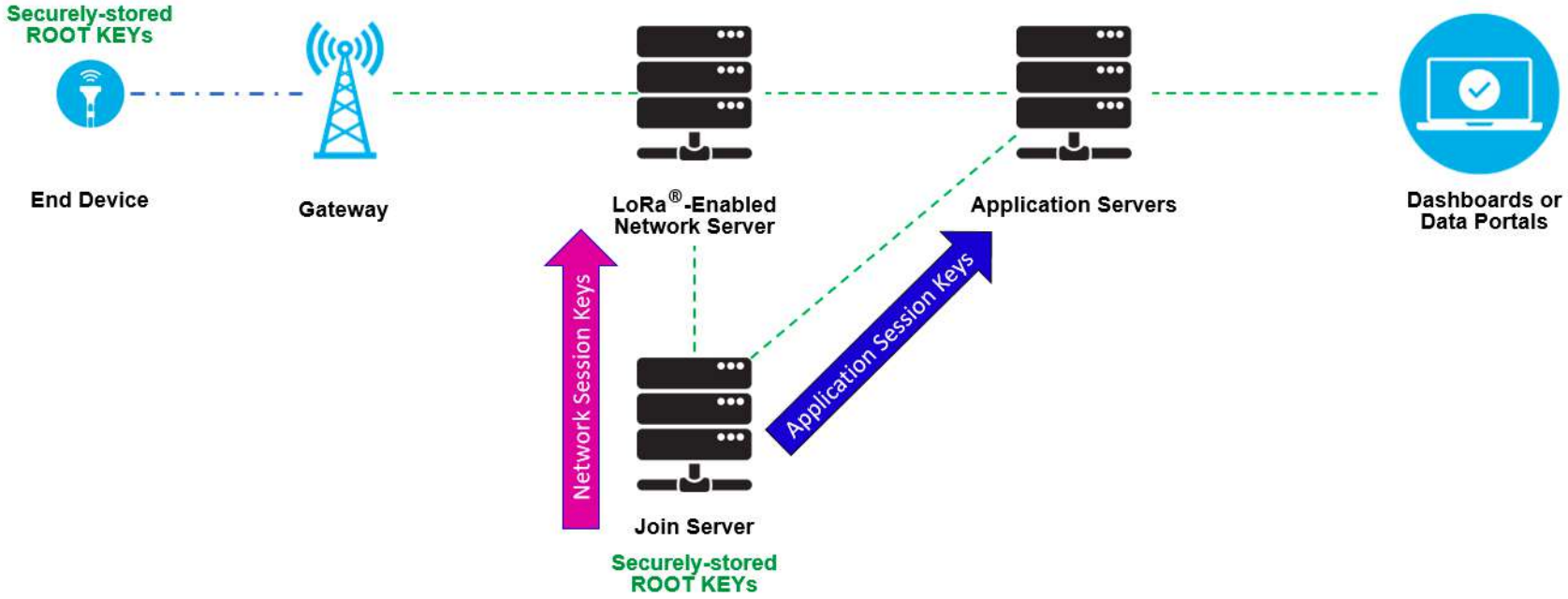
Join Response



source: https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

LoRaWAN Security: OTTA

Session keys distribution



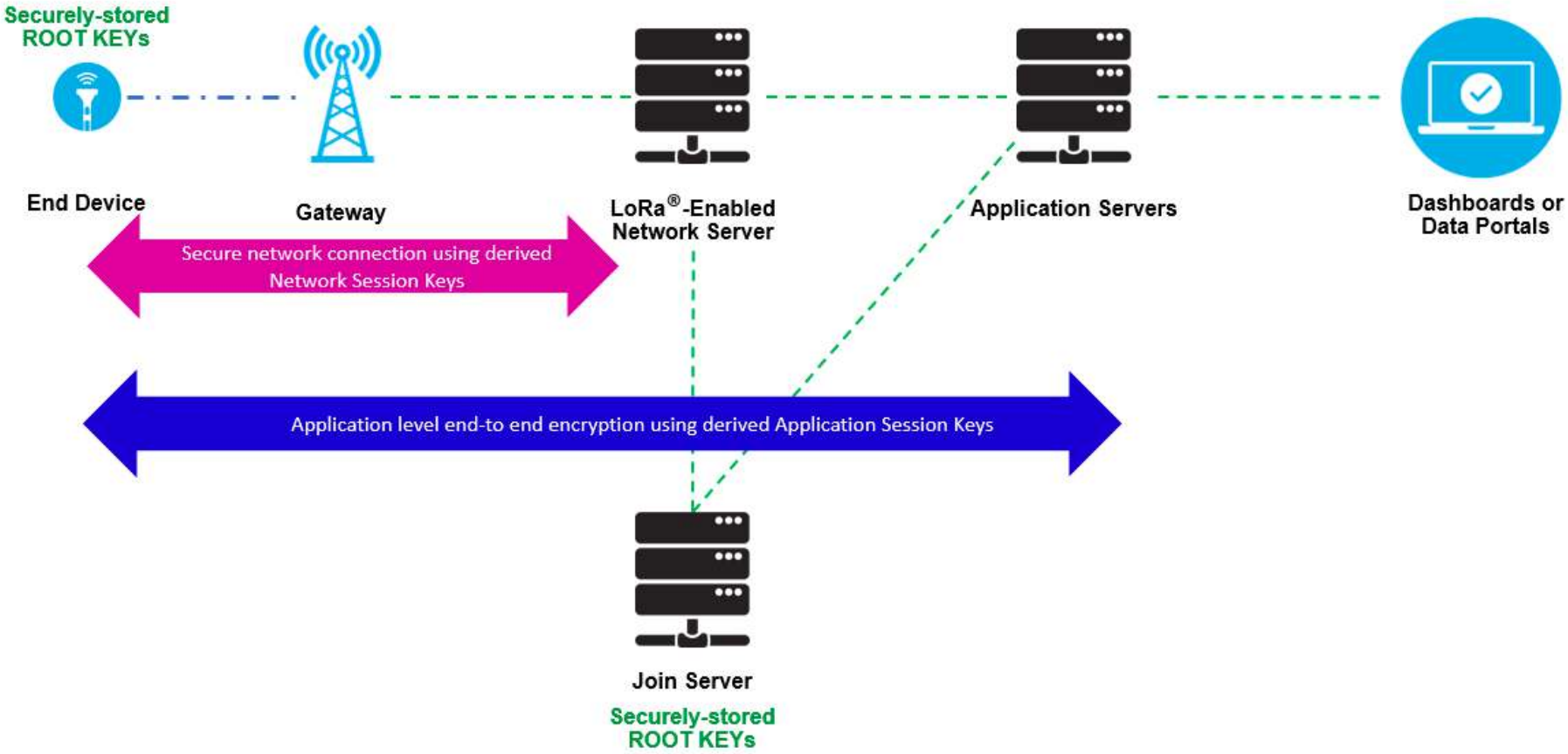
source: https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

LoRaWAN Security

LoRaWAN security: Two modes

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  - each end-node is provided with **NwkSKey and AppSKey**

LoRaWAN security: Summary



source: https://lora-developers.semtech.com/library/tech-papers-and-guides/lora-and-lorawan/

2- Sécurité de l'IOT

2.2 Réseaux Zigbee

Security of some technologies: 802.15.4/Zigbee

- IEEE 802.15.4 définit les fonctionnalités de sécurité (algorithme de chiffrement / intégrité, etc.) taille des MIC, etc. et les met à disposition des couches supérieurs.

- La gestion des clés est faite au niveau des couches supérieures (ex. Zigbee)

- L'algorithme de chiffrement utilisé est AES (Advanced Encryption Standard) avec une longueur de clé de 128 bits

802.15.4 Frame

Bits: 0–2	3	4	5	6	7–9	10–11	12–13	14–15
Frame Type	Security Enabled	Frame Pending	AR	PAN ID Compression	Reserved	Dest. Addressing Mode	Frame Version	Source Addressing Mode

Octets: 2	1	0/2	0/2/8	0/2	0/2/8	0/5/6/10/14	variable	2
Frame Control	Sequence Number	Destination PAN Identifier	Destination Address	Source PAN Identifier	Source Address	Auxiliary Security Header	Frame Payload	FCS
		Addressing fields						
MHR							MAC Payload	MFR

127 octets max

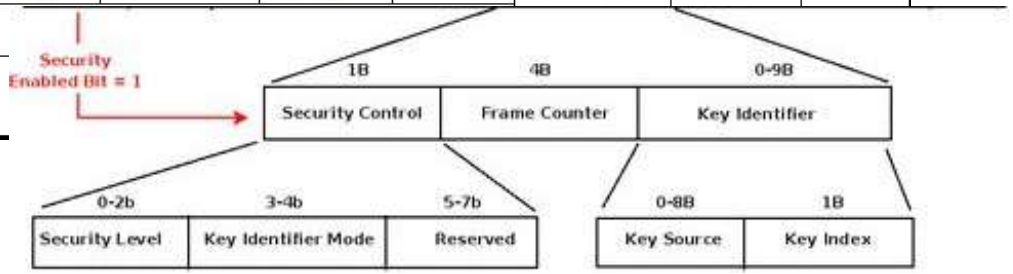
Source de la figure : Lopez, Javier & Rios, Ruben & Bao, Feng & Wang, Guilin. (2017). Evolving privacy: From sensors to the Internet of Things. Future Generation Computer Systems. 75.

More details : https://core.ac.uk/download/pdf/14699804.pdf and https://scdn.rohde-schwarz.com/ur/pws/dl_downloads/dl_application/application_notes/1gp105/1GP105_1E_Generation_of_IEEE_802154_Signals.pdf

802.15.4 Frame

Bits: 0–2	3	4	5	6	7–9	10–11	12–13	14–15
Frame Type	Security Enabled	Frame Pending	AR	PAN ID Compression	Reserved	Dest. Addressing Mode	Frame Version	Source Addressing Mode

Octets: 2	1	0/2	0/2/8	0/2	0/2/8	0/5/6/10/14	variable	2
Frame Control	Sequence Number	Destination PAN Identifier	Destination Address	Source PAN Identifier	Source Address	Auxiliary Security Header	Frame Payload	FCS



802.15.4 Frame: Security level

0x00	No security			Data is not encrypted. Data authenticity is not validated.	
0x01	AES-CBC-MAC-32		MIC-32	Data is not encrypted. Data authenticity is validated.	
0x02	AES-CBC-MAC-64		MIC-64	Data is not encrypted. Data authenticity is validated.	
0x03	AES-CBC-MAC-128		MIC-128	Data is not encrypted. Data authenticity is validated.	
0x04	AES-CTR		ENC	Data is encrypted. Data authenticity is not validated.	
0x05	AES-CCM-32		AES-CCM-32	Data is encrypted. Data authenticity is validated.	
0x06	AES-CCM-64		AES-CCM-64	Data is encrypted. Data authenticity is validated.	
0x07	AES-CCM-128		AES-CCM-128	Data is encrypted. Data authenticity is validated.	

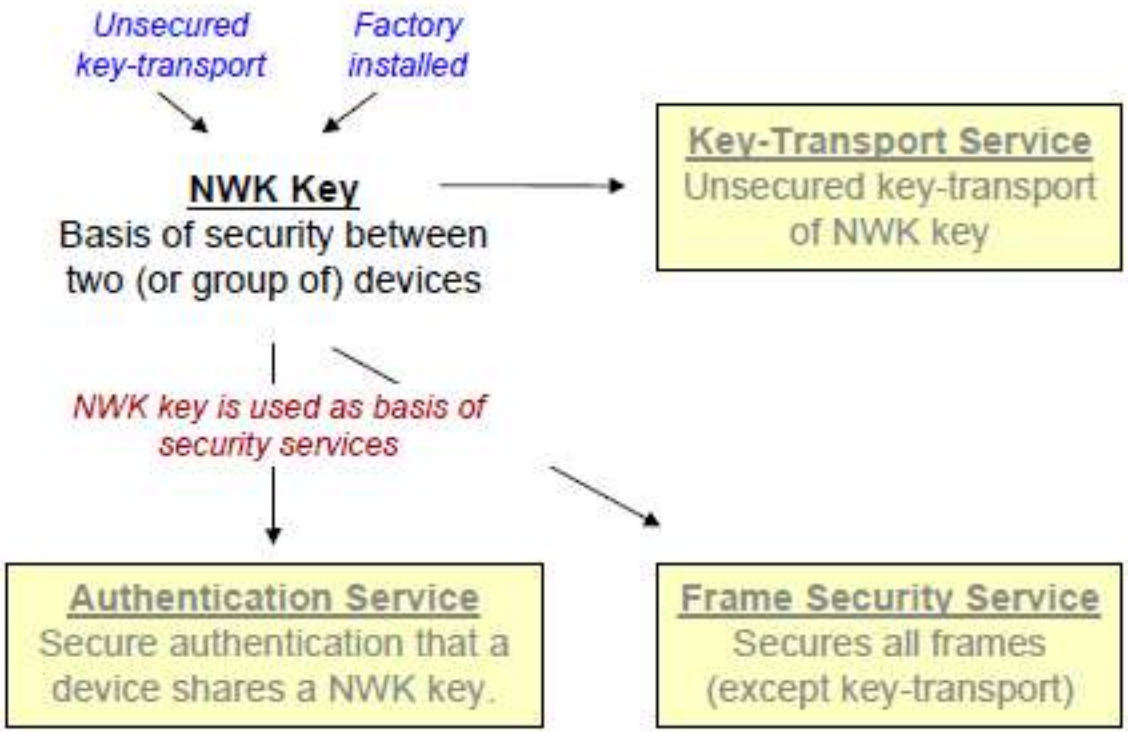
Security of some technologies: Zigbee

- **Zigbee: security policies**
  - **Residential mode:** Network Key only
  - **Commercial mode** : Master key and Link keys

- **Zigbee Residential mode : (Network key)**
  - A unique 128b key shared among all the devices in the network
  - Pre-installation or Key-transport
  - Each router decrypt and verify each packet before forwarding it

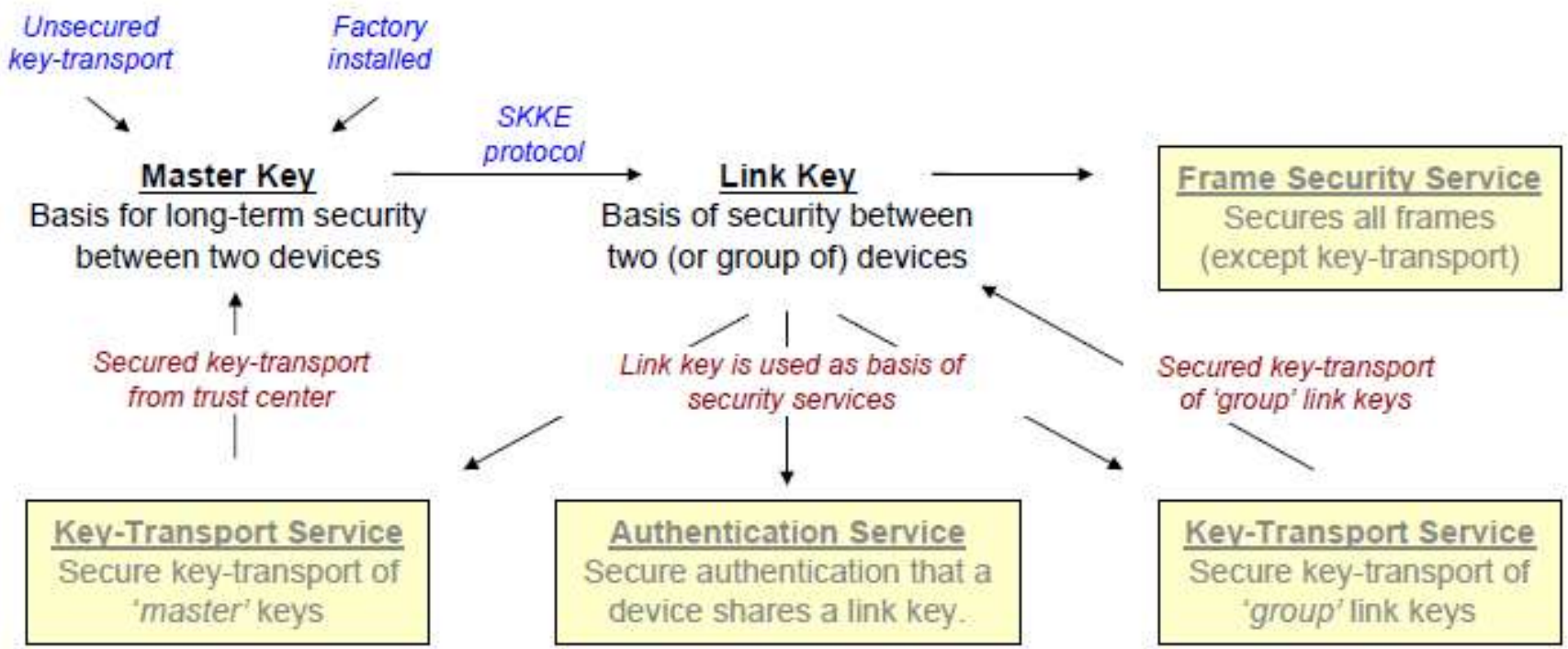
- **Zigbee Commercial mode : (Master and link keys)**
  - Master Keys: pre-installed or Key-transport
  - used for Key Establishment Procedure (SKKE) to establish link keys.
  - Link key: unique between each pair of nodes

Security of some technologies: Zigbee



ZigBee Residential Mode: source https://sbt-alliance.com/wp-content/uploads/2020/04/Daintree-DT200-GE-Zigbee-Primer-Whitepaper.pdf

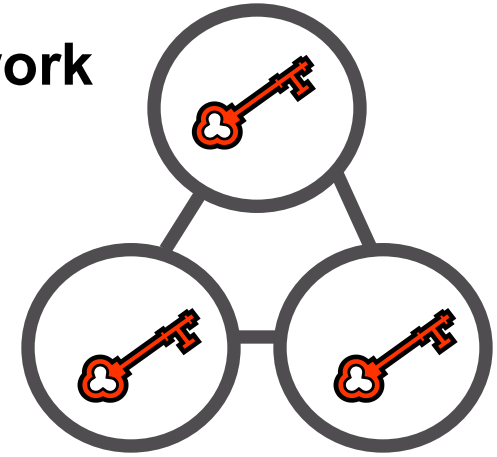
Security of some technologies: Zigbee



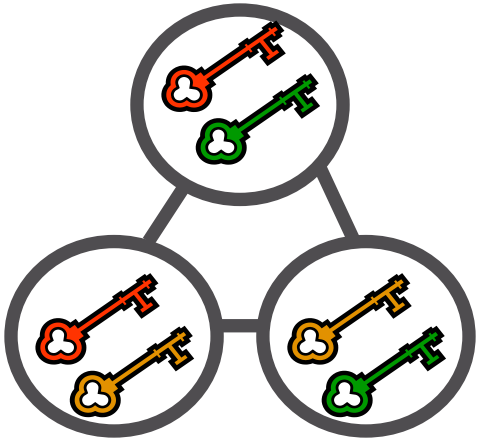
ZigBee Commercial Mode: source https://sbt-alliance.com/wp-content/uploads/2020/04/Daintree-DT200-GE-Zigbee-Primer-Whitepaper.pdf

Unique key Vs link keys

- **A unique key to secure all the network**
  - –Memory efficiency
  - –Very bad network resiliency



- **One Key by link**
  - –Perfect resiliency
  - –Huge memory requirement



2- Sécurité de l'IOT

2.3 Ouverture sur des problèmes de recherche

Key management in Ad-Hoc WSN : Symmetric Vs Asymmetric

Symmetric cryptosystems

- Simplicity
- Fast computation
- Small size of required keys
- Symmetric key agreement

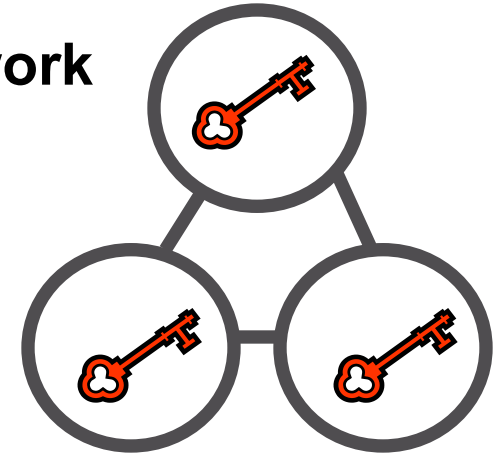
Asymmetric cryptosystems

- Efficient key management
- Heavy computations
- Large size keys
- Complex and slower operations

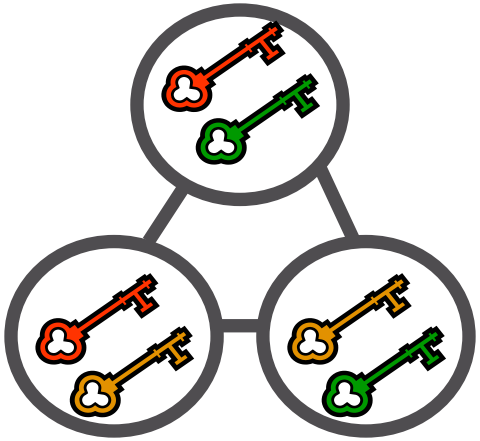
Resource limitations (energy, memory and computations)	$\longrightarrow$	Symmetric key management
Lack of infrastructure	$\longrightarrow$	Key pre-distribution

Unique key Vs link keys

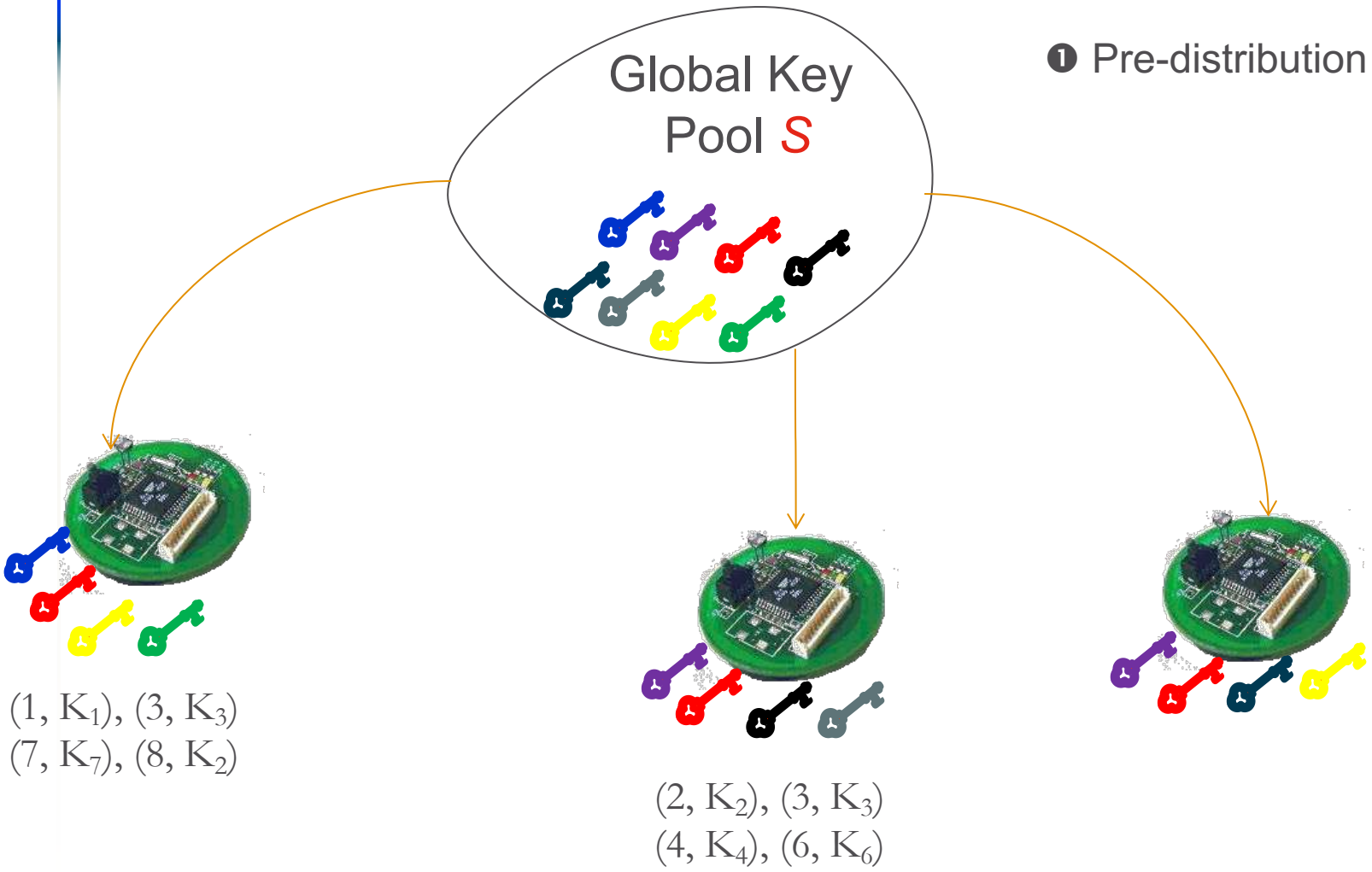
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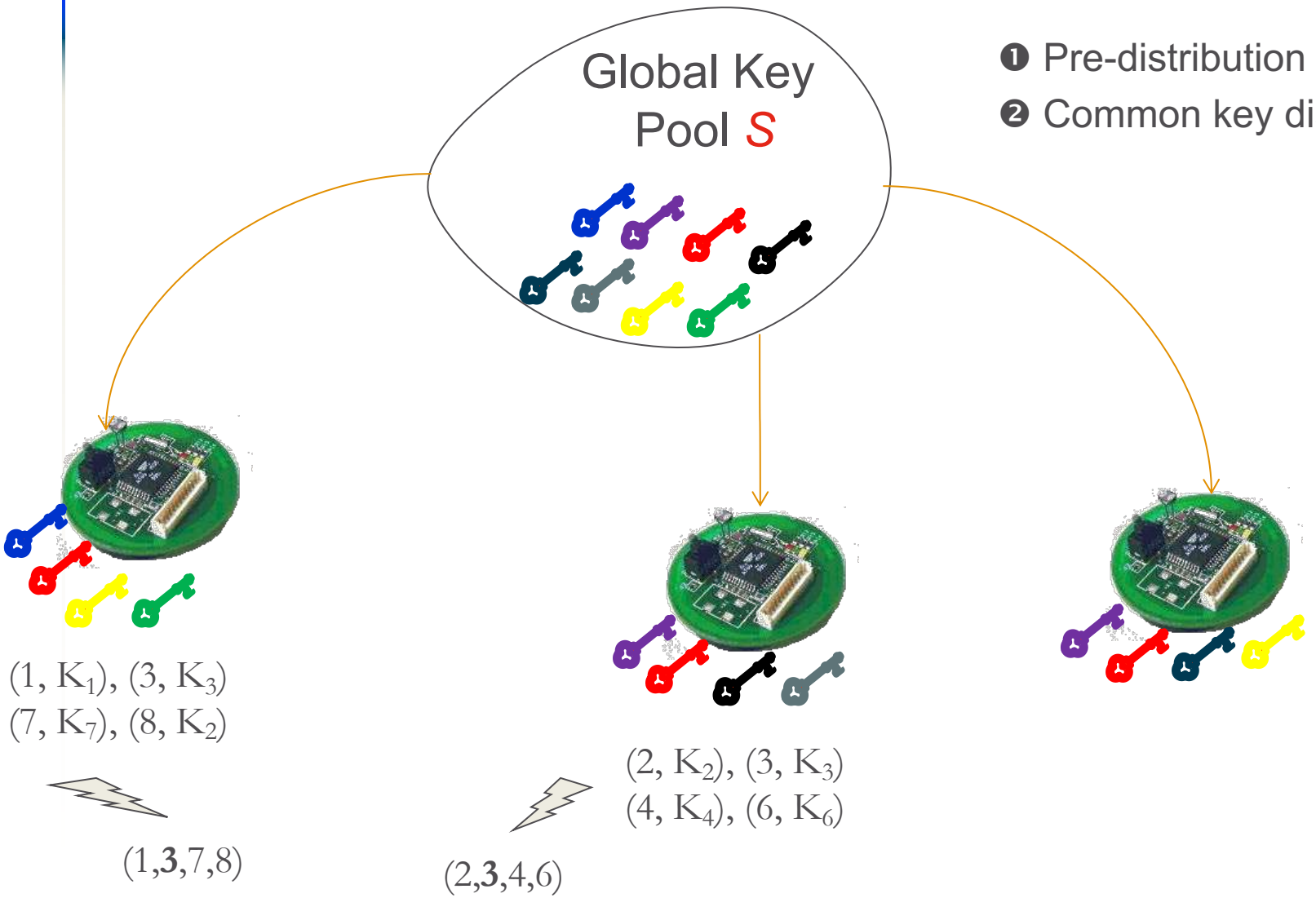
- **One Key by link**
  - –Perfect resiliency
  - –Huge memory requirement



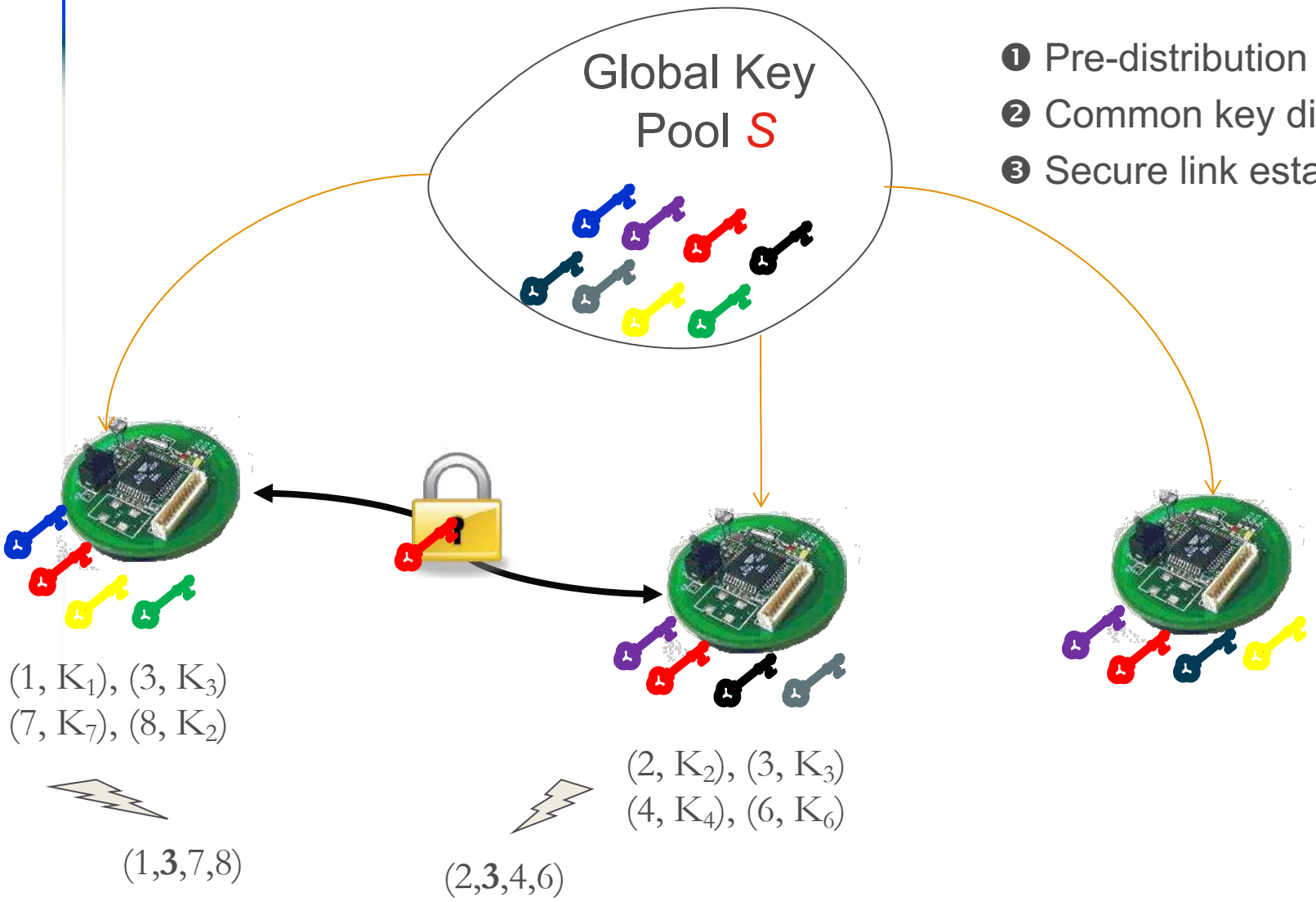
Key pre-distribution for Ad-Hoc WSN



Key pre-distribution for Ad-Hoc WSN



Key pre-distribution for Ad-Hoc WSN



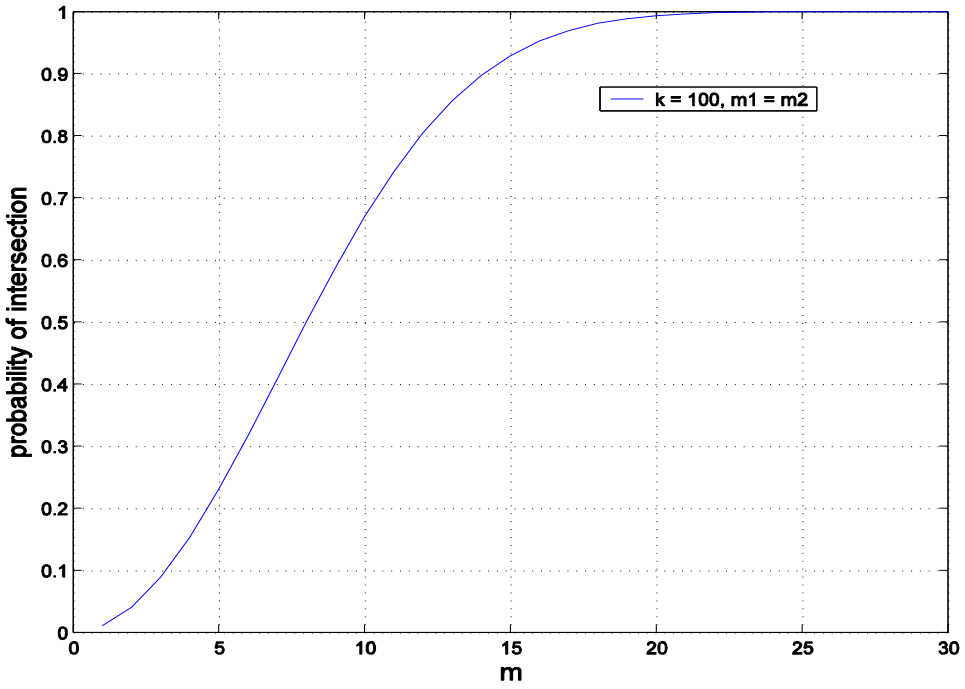
Evaluation Metrics

- **Key ring size :** Storage overhead
- **Key sharing probability (network secure connectivity) :**
  probabilistic schemes Vs deterministic schemes
- **Network resiliency :** fraction of uncompromised external links when nodes are compromised
- **Network scalability**
- **Communication overhead**
- **Computation overhead**

Connectivity of the basic scheme

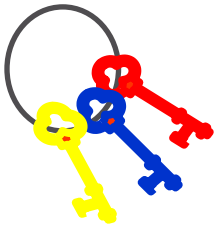
Given a set S of k elements, we randomly choose two subsets S_1 and S_2 of m_1 and m_2 elements, respectively, from S.

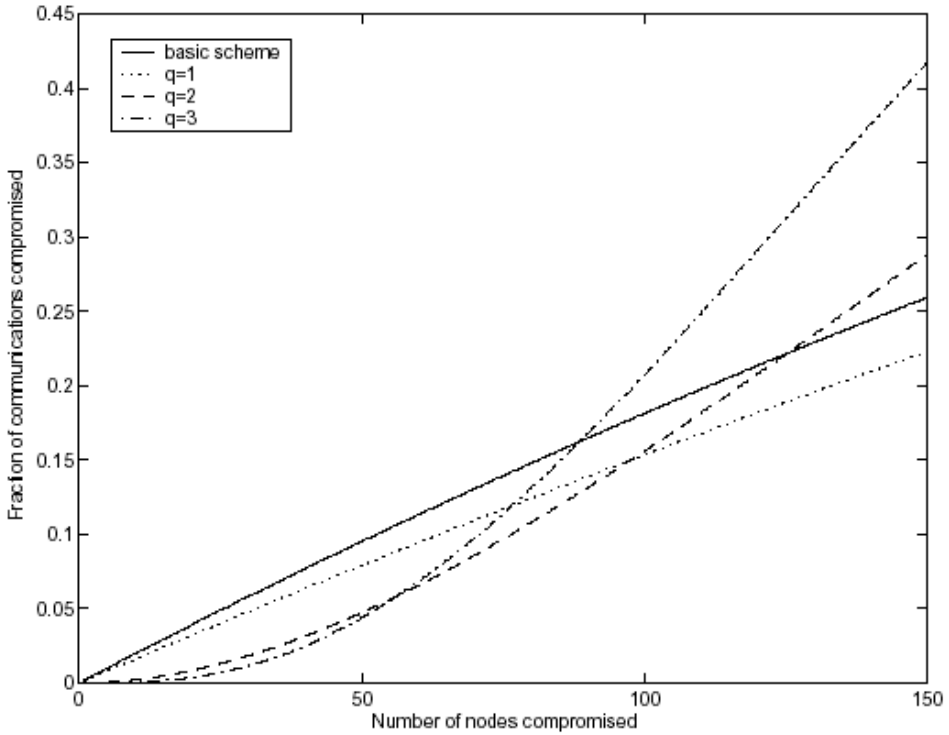
$$\Pr\{S_1 \cap S_2 \neq \emptyset\} = 1 - \frac{(k-m_1)!(k-m_2)!}{k!(k-m_1-m_2)!}$$



Q-Composite Scheme

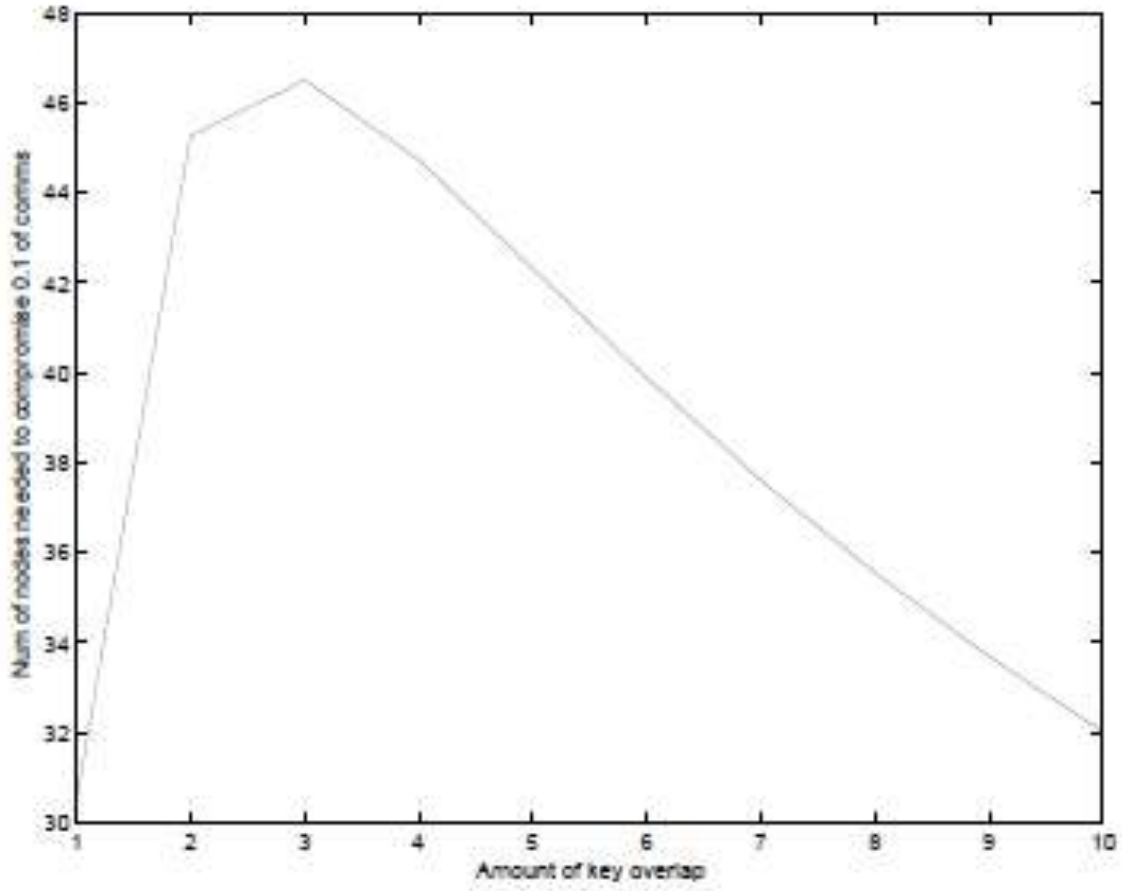
- At least Q common keys to establish a secure link
- The session key is a combination of all common keys





Fraction of communications compromised k = 200, p = 0.33

Q-Composite Scheme



nodes needed to compromise 0.1 of comms (k = 200, p = 0.5)

Some key management schemes for WSN

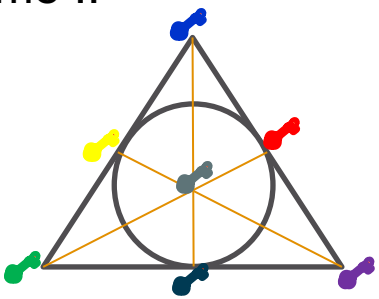
- **Probabilistic schemes**



  – Q-composite scheme

  – Deployment knowledge based scheme, Polynomial based schemes, BK scheme, [Ruj et al. 2011] Trade Based scheme ..

- **Deterministic schemes**



–Symmetric Balanced Incomplete Design based scheme. Ex: {1,2,3} {1,4,5} {1,6,7} {2,4,6} {2,5,7} {3, 4, 7} {3, 5, 6}

– Key pre-distribution using Partially Balanced Incomplete Block Design

Thank you !